Design and Demonstration of Integrated Silicon Devices for Optical Sensing and Optical Communications

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PhD. Thesis

December 2020

A thesis submitted to Khalifa University of Science and Technology in accordance with the requirements of the degree of PhD in Interdisciplinary Engineering in the Department of Electrical Engineering and Computer Science.

Design and Demonstration of Integrated Silicon Devices for Optical Sensing and Optical Communications

by

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A thesis submitted in partial fulfillment of the requirements for the degree of

PhD in Interdisciplinary Engineering

at

Khalifa University

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December 2020

Abstract

Ayat Magdy Taha, "**Design and Demonstration of Integrated Silicon Devices for Optical Sensing and Optical Communications**", PhD Thesis; PhD in Interdisciplinary Engineering, Department of Electrical Engineering and Computer Science, Khalifa University of Science and Technology, United Arab Emirates, December 2020.

In recent years, gas sensors at a micro-and-nano scale have gained a lot of attention especially in critical areas such as environmental and medical fields. The development of such sensors is accompanied by advances in micro-and-nano fabrication as well as through exploration of new materials, which has led to enhanced gas sensing performance. The focus is to achieve a real-time, highly sensitive and selective integrated gas sensors while overcoming the fundamental challenges such as device footprint, cost and temperature fluctuations that influence the sensor's behavior.

The presented work is a theoretical and experimental analysis of different optical gas sensors based on the Mach-Zehnder interferometer (MZI) design, created on silicon-on-insulator platform. The proposed sensors have enhanced the possibility of gas detection with high sensitivity and low detection limit in the optical domain. The sensors are susceptible to the undesired effect of temperature change which originates from the subjection of different gases. Therefore, a detailed description on a simple approach to counteract this change via the use of microring resonator is introduced. Two MZI configurations were studied: a partially exposed and a suspended interferometer, for which the estimated sensitivities were 2071 nm/RIU and 5500 nm/RIU, respectively. The sensing experiment was conducted using: Helium and Nitrogen gases.

Finally, an integrated optical readout system is introduced via the use of Arrayed waveguide grating (AWG) and Multi-mode Interferometers (MMIs). The integrated spectrometer

eliminates the need for expensive off-chip bulky systems. AWG devices are typically used for wavelength division multiplexing (WDM) in the optical telecommunication field. However, the same concept may also apply for spectroscopic intents. In this work a 4×4 MMI was designed and fabricated with efforts directed to secure a 100 GHz channel spacing in 4-channel WDM, a 3-dB optical bandwidth of about 100 GHz, high uniformity and excess loss less than 1 dB for the MMI couplers.

Indexing Terms: Silicon-on-Insulator; Optical Gas Sensor; Microring Resonator; Integrated Spectrometer.

Contribution to Khalifa University and Relevance to the UAE

Most of the research carried out at KHALIFA University-Masdar campus, under the Nano-Optics and Optoelectronics Research (NOOR) Laboratory, aims to serve in qualifying microsystems technologies established within the emerging semiconductor industry in the United Arab Emirates (UAE). The research is rather increasingly evolving especially via the introduction of new ideas that take advantage of optical devices in distinct applications on an SOI platform. The current work is particularly focused on gas sensing applications in which efforts for achieving high sensitivities and low limits of detection are made. The sensor is intended to detect various atmospheric gases, nontoxic and toxic gases with enhanced sensor calibration. Gas sensing technology has gained a lot of interest in both, industry and academia to satisfy the growing market demands. The technology is peculiarly important due to its widespread application in areas such as: industrial production, medical, and air quality monitoring, etc. Gas sensing by means of optical methods in the macro-to the micro-scale is very promising due to the achievement of high sensitivities, and shorter response time which enables real time detection. Silicon photonics market has a rising study that investigates deeply into the development and commercialization of eco-efficient products. Similarly, KHALIFA University and the UAE revolve around the same endeavors in the field of healthcare, agriculture, and environmental monitoring. Therefore, this current work aligns with the essential requirements for tackling some of the intricate issues by extending the application of optical technologies in these fields.

Acknowledgement

I would first like to thank God Almighty for helping me finish this PhD program with great success and enriching me with all the strength and tenacity I needed during my journey. I would then like to thank my lovely family: my Mom- Fatima Hanem, my Dad- Magdy Taha, my Siblings and especially my twin Inas Magdy, for their immense support and prayers, and for highlighting the significance of education and by all means for their plentitude love and understanding.

I would like to thank my primary advisor Dr. Marcus Dahlem for all the support and guidance he offered during the first year and a half of my PhD. I would like to expressly thank my second advisor Dr. Jaime Viegas for giving me the opportunity to join his skilled and humble group members and for his enormous support especially during the rough year of 2020. I also want to thank my research advisory committee: Dr. Mihai Sanduleanu and Prof. Yoshi Nakano for their support and help during my research journey.

I want to thank Mr. Leslie George, Dr. Cyril Aubry, and Dr. Florent Ravaux for all the lab trainings that enhanced my skills and helped me in my research work.

I want to give special thanks to all my group members and to my very dear friends: Dr. Badreyya Al Shehhi, Dr. Raquel Flores, Dr. Ricardo Janeiro, Dr. Soha Yousof, Dr. Sam Sofela and Bussy, Bruna Patricia, Saeed Ali Al Tenaji, Dr. Hossam Shoman, Hayk Gevorgyan, Dr. Salim AlKaabi, Dr. Dan Cracan, Juan Villegas, Mutasim Odeh, Jose Gerardo, Sohail Shah, Bahawal Haq, Mervet Al Shehhi, Dr. Solomon Serunjogi, Dr. Tarek Alkhatib, Mustafa Safadi, Thaer Alafghani, Jehad Abed, Khaloud Alnaimi, Mavy Pais, Elangovan, Chukwuka Madumelu, Dr. Oraib Ghaith and Dr. Karen Sloyan for being part of my journey, for being my

mentors, my stress relief and for all their support, love, encouragement and for being my second family.

Finally, to Khalifa University, I am greatly thankful for being given the chance to complete my PhD program under your granted scholarship.

This thesis is particularly dedicated to my family and my niece Jory Wael.

Ayat Magdy Taha

Khalifa University (Masdar Institute), December, 2020

Declaration and Copyright

Declaration

I declare that the work in this thesis was carried out in accordance with the regulations of Khalifa University of Science and Technology. The work is entirely my own except where indicated by special reference in the text. Any views expressed in the thesis are those of the author and in no way represent those of Khalifa University of Science and Technology. No part of the thesis has been presented to any other university for any degree.

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Nomenclature

MZI Mach-Zehnder Interferometer

AWG Arrayed Waveguide Grating

MMIs Multi-Mode Interferometers

WDM Wavelength Division Multiplexing

SOI Silicon-on-Insulator

PICs Photonic Integrated Circuits

GVD Group Velocity Dispersion

UV Ultraviolet Lithography

EBL Electron Beam Lithography

MRR Microring Resonator

BGW Bragg Grating Waveguide

FSR Free Spectral Range

LOD Limit of Detection

SWG Sub-Wavelength Grating

RAMZI Ring-assisted MZI

DC Directional Coupler

FDTD Finite Difference Time-Domain

PML Perfect Matching Layer

FDE Finite-Difference Eigenmode

SPS S-Parameter Simulator

TO Thermo-Optic

SPAR Optical N-Port S-parameter

FPR Free Propagation Region

DeMux Wavelength Demultiplexer

SDM Spatial-Division Multiplexing

MDM Mode Division Multiplexing

MFC Mass Flow Controller

N2 Nitrogen

He Helium

PCR Polymerase chain reaction

PMUTs Piezoelectric Micromachined Ultrasonic Transducers

MPW Multi-Project Wafer

PDMS Polydimethylsiloxane

Chapter 1: Introduction

1.1 Motivation

The work in this dissertation investigates optical sensors, with specific emphasis on prominent optical devices like the Mach-Zehnder interferometer and the microring resonator. As indicated earlier, silicon photonics has paved its way into the sensing domain to contribute in several applications like in medicine, food quality, and the environment, to name a few. Its contribution to biosensing and gas detection applications is especially overwhelming in comparison to existing technologies. Therefore, the work presented here builds upon different designs for silicon photonic sensors and demonstrates their role in environmental sectors which are especially pertinent and widely used in the sensing industry. Moreover, integrating several laboratory functions on a single chip is of significant importance. A dispersive element which successfully spatially segregates the light into its constituent wavelengths without the need for an external spectrometer is hence, also studied. To this end, the work presented here extends from presenting a simple sensing functionality to an integrated readout system designed for improved performance, without an external instrument for quantifying results.

1.2 Silicon Photonics

Various applications in microelectronics require an exceptionally influential technology platform which is based on silicon chips. This platform, which has been developed over the last decades at an increased rate, is now making discoveries in science and technology in all parts of the world owing to the sharp increase in investment by industry and universities. It has been demonstrated that with the

introduction of the silicon-on-insulator (SOI) technology, silicon based photonic integrated circuits (PICs) have become potential [1]. Another prominent aspect of an SOI structure is the high refractive index contrast between the silicon core (n≈3.45) and SiO_2 cladding (n≈1.45) which yields strong optical confinement [2, 3]. This feature facilitates tight bends with radius down to few micrometers, enabling closely packed devices and thus high density integration of optical devices on a chip [4, 5]. The blend of these advantages together with the ease of fabrication via a well-established conventional silicon microelectronics processing has opted SOI integrated optical circuits in optical sensor and communication applications [6]. Furthermore, SOI platform possess a factual compatibility with silicon CMOS integrated circuit technology, which makes it a promising candidate for future, less costly photonic integrated circuits [7]. CMOS circuits that were fabricated on an SOI platform have gained an advantage over reduced parasitic effect and no latch-up which thereby enables a high-speed operation at a reduced power. These are the vital requirements in several telecom and computation systems [8].

All through the emerging SOI technology, several different SOI guided wave optical devices have been revealed. Integrated 3-dB directional couplers [4], low loss multimode interference (MMI) couplers [9], asymmetric Mach–Zehnder wavelength filters [7], SOI ring resonator sensor [10], and Mach-Zehnder interferometer (MZI) sensor [11] are some of the SOI optical devices that have been demonstrated over the past few years. Figure 1-1 depicts a monolithically integrated PIC in a single material platform[12].

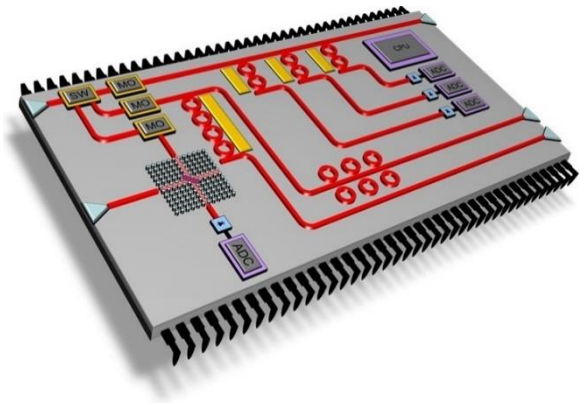


Figure 1-1. Typical integrated chip which includes both electronic and photonic devices. Devices included are optical waveguides, light-chip couplers, splitters and combiners, filters and photodetectors [12].

Waveguides are optical components that aim in guiding light from one place on a chip to another with low propagation loss, small feature size and more compact bend radius for realizing high density chips [13, 14]. Optical waveguides are usually classified as per their geometry (Figure 1-2) for instance rib or strip waveguide, their mode profile: single-mode or multimode (Figure 1-3), polarization: transverse electric (TE) or transverse magnetic (TM) (Figure 1-4) and their material [14]. It is worth stating that in most cases, depending on the application, single-mode operation is favored over multimode [15, 16]. In multimode waveguide, the higher order modes propagate with a propagation constant (β) value smaller than that of the lower order modes and are less confined in the waveguides [17-19]. Propagation constant is a figure that determines how the amplitude and the phase of light at a given wavelength changes along the direction of propagation. Therefore, having this offset in phase between the higher and lower order modes will result in a modal dispersion and the reduced optical confinement will induce unpredictable losses due to a large field decay into the cladding [1].

Waveguide modes are electromagnetic field patterns that travel along the longitudinal coordinate of the waveguide, with a distinguishable cross-sectional intensity profile, phase velocity, group velocity, and polarization [20]. In the case of transverse electric (TE) mode, the electric field is dominant in the horizontal direction, while in the case of transverse magnetic (TM) mode, the electric field is dominant in the vertical direction as shown in Figure 1-4. Each polarization will suffer from different losses depending on their profile and how they overlap with the waveguide sidewall as they exhibit different field discontinuities [1, 21, 22]. It is apparent from Figure 1-4 that TE mode experiences field discontinuities at the high index contrast waveguide sidewalls. Therefore, TE mode suffers more from the sidewall roughness loss as compared to TM mode where the field discontinuity takes place perpendicular to the direction of propagation. Inevitable surface roughness is the fundamental reason behind waveguide propagation loss as well as scattering [23-25]. TM modes on the other hand, are more delocalized from the waveguide, this makes them more susceptible to substrate radiation loss, notably in the waveguide bend region. Waveguide bend loss is originated when the effective index of the waveguide goes below the refractive index of the cladding layer, an effect which is prominent in small bend radius. However, given that SOI waveguides have a high index contrast between the Si core and the SiO_2 cladding, thus far, SOI waveguides permits the sharpest bends in any present integrated waveguide technology [18, 22].

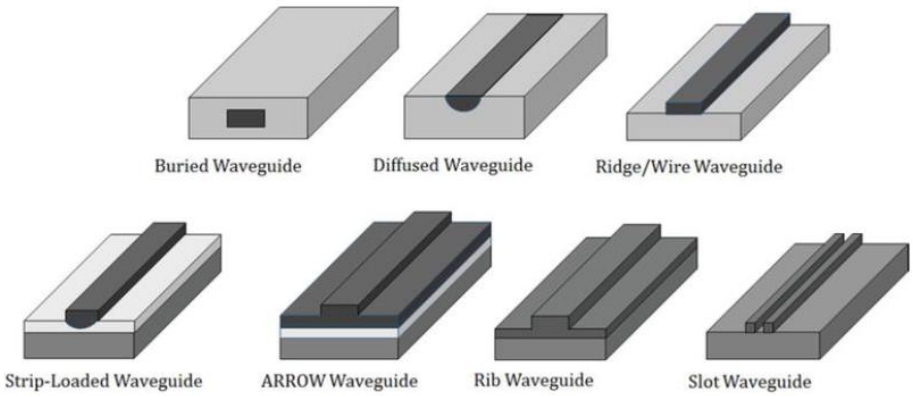


Figure 1-2: Basic schematic of different types of waveguides. Typical dimension for a single-mode operation is 450x220 μm^2 [20].

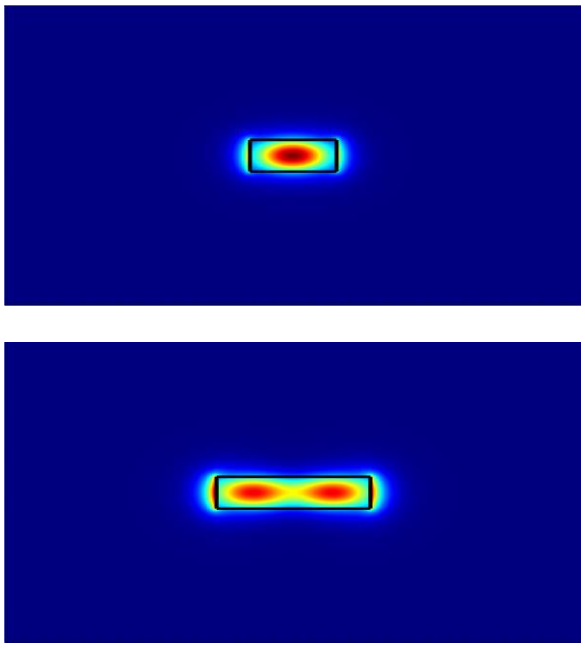


Figure 1-3: Modal field profiles for the optical buried waveguide; (a) Single mode, and (b) Multimode.

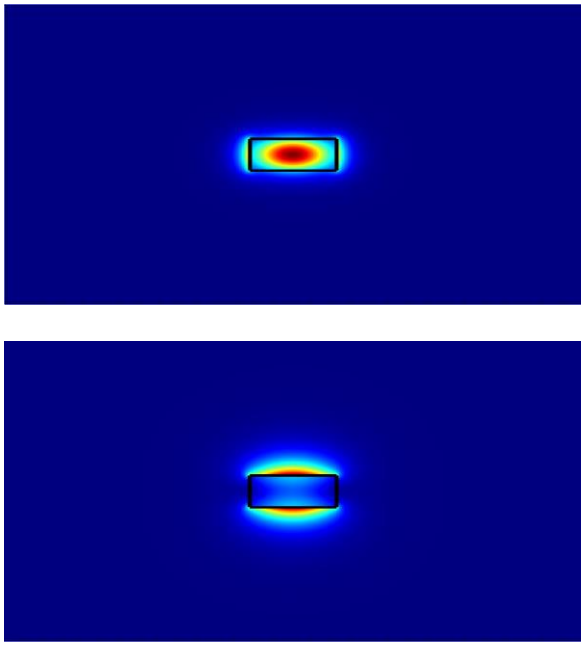


Figure 1-4: Optical mode profiles for a typical buried waveguide for; (a) TE mode at λ=1550 nm, and (b) TM mode at λ=1550 nm.

An effective way to deal with these losses is, for the case of TE polarization, to increase the waveguide width wide enough to avoid the modal overlap with the sidewalls given that TE modes are more sensitive to the sidewall roughness. As for TM polarization, increasing the bend radius would inevitably reduce the mode spread into the cladding and consequently reduce the waveguide bending losses. Another preventable loss for both TE and TM polarizations is the substrate leakage loss. Despite the fact that in SOI waveguide, the modes are well confined, the exponential tail (evanescent tail) of the guided mode (Figure 1-5) stretches out of the cladding into the substrate, resulting in a substrate leakage loss. TM mode is distinctly more affected by the substrate leakage loss, due to its more delocalized profile. Increasing the cladding thickness would efficiently diminish the substrate leakage loss [26, 27].

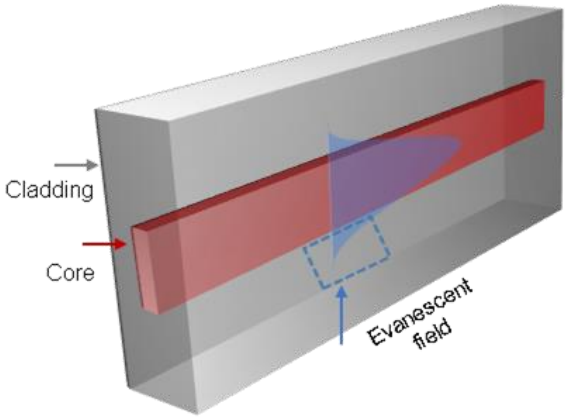


Figure 1-5: Image of the fundamental mode in a buried waveguide, revealing the evanescent field spreading into the cladding.

It is vital to mention that every waveguide mode, TE or TM, is categorized by means of different aspects. The most prevalent characteristic is the effective index which is denoted by n_{eff}, and the other is the propagation constant. Effective index is a parameter that gives details of the phase velocity of the light propagating in a medium in the same manner as the refractive index. Effective index in simple terms is the weighted average of the refractive index between the core and the cladding of the waveguide. Stronger optical confinement in the core of the waveguide renders a higher effective index. For a typical buried waveguide dimension of 450×220 nm, the effective index value at 1550 nm is usually around 2.35 and 1.75 for TE and TM mode, respectively. Optical waveguides with typical dimensions are dispersive, which implies that the effective index is wavelength dependent. This effect is predominantly due to the geometry of the waveguide in which the dimension of the core is on the same scale as that of the wavelength. Therefore, any minor change in the wavelength will alter the optical confinement and thus the effective index [28, 29]. On the other hand, group index, denoted by n_g, is a parameter defined by the ratio of speed of light

in vacuum to the group velocity in the medium. Group index is related to the effective index by the following expression [28, 30-32]:

$$n_g(\lambda) = n_{eff}(\lambda) - \lambda \frac{dn_{eff}}{d\lambda} \quad (1)$$

A wave packet of a pulse in a medium propagates at a speed identified as the group velocity, and it is the wave envelope which conveys the information. Group velocity dispersion (GVD) is a phenomenon that occurs because group velocity is wavelength dependent [32, 33]. As a result of dispersion, n_g is drastically higher than the effective index and in a typical buried waveguide it is around 4.3 for TE mode.

In silicon photonics technology, a vast number of environmental, scientific and commercial applications have been established. Some of which are, optical routers and signal processing, optical interconnects, long range telecommunications, and integrated optical sensors. Sensing applications which are based on optical devices have revealed a significant increase in sensitivities mainly due to the high index contrast feature. The high contrast enhances the sensitivity of the waveguide to a minor change in the cladding's refractive index, an effect which is particularly noticeable in TM mode than in TE mode [34]. Considering the fabrication of the optical devices, deep ultraviolet (UV) lithography is favored and usually the suited technique for large-scale mass fabrication. Furthermore, deep UV lithography is the standard technique used in CMOS fabrication, it has an advantage of high throughput when compared with other lithography techniques such as electron beam lithography (EBL). Despite the lower resolution, UV lithography resulted in low-loss waveguides in SOI platforms. For higher resolution EBL is rather preferred but is regarded as a very slow technique [26].

1.3 Scope of Thesis

The dissertation is concentrated on different silicon photonic sensor designs and their application in varying sensing sectors, as well as their integration with an on-chip spectrometer for a more compact sensing platform. The following points presents a synopsis of every chapter in this thesis.

This chapter offers an overview on the state of art of silicon-on-insulator technology and silicon based photonic integrated circuits. A brief description of optical waveguides and optical modes is also presented.

Chapter 2 reviews the theme, trends, and challenges in silicon photonic sensor designs and applications. It offers synthesis of the existing sensing technologies with focus on the contribution of literature to gas and bio-sensing applications. The various optical topologies that are discussed demonstrate a comprehensive understanding of optical sensors and their functionality.

Chapter 3 introduces the common modeling and numerical methods such as Finite Difference Time-Domain, Mode Solver, Lumerical INTERCONNECT, and S-Parameters for waveguide design and sensor optimization.

Chapter 4 focuses on the design and operation of the proposed structures: partially exposed and suspended Mach-Zehnder interferometers. The chapter also demonstrates the potential of a temperature compensating ring resonator for accurate sensing quantification and eliminating ambiguous detection. The concept of the integrated spectrometer using arrayed waveguide grating and multimode interferometer is also proposed in this chapter of the thesis.

Chapter 5 demonstrates the performance of the fabricated sensors that were optically characterized as well as measured in a gaseous medium using nitrogen and helium

gases and in a liquid medium using DI water and IPA. The temperature sensitivity of the proposed sensor designs is also determined and presented in this chapter. A clear description of the characterization results of the arrayed waveguide grating-based spectrometer is also presented.

Chapter 6 summarizes and concludes the simulation and experimental results acquired in the preceding chapters and conveys perspectives for significant and potential future research in the area of silicon photonic sensors.

Chapter 2: Compact Silicon Photonic Sensors

This chapter will briefly highlight on the different types of optical components used in sensing applications. The optical sensing components can be classified as intrinsic or extrinsic sensors, where the former is more emphasized on in this chapter. An optical sensor serves as a transducer in which the measurand changes a property in the light beam which is then translated into a useful measurable output signal therefore, optical sensors are categorized based on the phenomena that is being measured.

2.1 Integrated Optical Sensors

There is an emerging interest in the field of optical sensing owing to their numerous benefits, such as low power consumption, high sensitivity, low cost, immunity to electromagnetic interference, robustness, and their capability of being integrated with electronic devices [35]. These advantages necessitates the use of optical sensors in various applications including environmental, industrial and medical [36]. However, optical sensors do suffer from some drawbacks, which are usually very challenging to mitigate such as poor selectivity, in some cases, and perturbations from undesired factors. In the former, optical sensors strive to distinguish different analytes within the same environment, while in the latter, a common complication is to completely eliminate the temperature dependency in most optical sensing components. Another disadvantage is that the ambient light can effectively interfere with the function of the sensor [37].

There are two distinct types of optical sensors: intrinsic and extrinsic sensors whereby, in intrinsic sensors the optical waveguides interact promptly with the measurand resulting in a variation in a particular optical property of the light. The measured

changes in the light property can be intensity, phase, polarization, and wavelength which are a direct consequent of a change in the waveguide parameters such as the optical length or the refractive index, manifested in the presence of the measurand [35, 38]. In contrast, extrinsic sensors do not have a direct interaction between the measurand and the optical waveguide and are typically more costly and bigger than the intrinsic sensors. Therefore, considering that in intrinsic sensors the optical waveguide plays the role of a transducer, it is of great importance to carefully design them by following a comprehensive strategy to ensure a high sensing performance. An additional important parameter in optical sensors is the operating wavelength given that individual elements exhibit a different absorption spectra in the mid-IR region. Accordingly, propagation at mid-IR wavelengths would be encouraged in order to intensify the function of the sensor by implementing homogeneous sensing as well as optical absorption [35]. This is particularly important for effective monitoring of toxic gases such as (CO_2), carbon monoxide (CO), methane (CH_4), and sulfur dioxide (SO_2). The detection of these gases is based on the principle of absorption peaks near the mid-infrared spectrum where the absorption intensity gives information about the concentration of the harmful gas [39, 40].

There are numerous different types of integrated optical sensors out of which Microring Resonator (MRR)-based sensors, Multimode Interferometers (MMI)-based sensors, Bragg Grating Waveguide (BGW)-based sensors, and Mach-Zehnder Interferometer (MZI)-based sensors will be summarized in the following chapter.

2.1.1 Microring Resonator based sensors

MRRs are optical components comprised of an optical waveguide twisted back onto itself and a set of parallel bus waveguides, namely the through and the drop waveguides as shown in Figure 2-1. Light is evanescently coupled between the looped

waveguide and the bus waveguides and interferes constructively when the resonance condition is met. The resonance condition is satisfied when:

$$L = 2\pi R = m\frac{\lambda_{res}}{n_{eff}} \tag{2}$$

The perimeter of the ring is denoted by L, R is the radius of the ring, m is the resonance mode order, n_{eff} is the effective refractive index, and λ_{res} is the resonance wavelength which fulfills the resonance condition [41]. In what follows, we will give a brief description on some of the concepts and formulas which are essential to set the criteria for the design of ring resonator sensors and for computing their functional behavior.

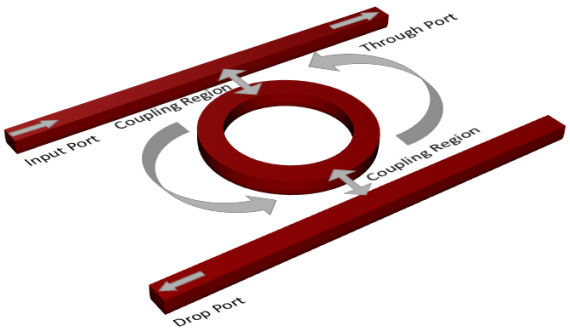


Figure 2-1: Schematic illustration of a microring resonator.

The period or the free spectral range (FSR) of the resonances of the microring resonator as a function of wavelength is given by the following expression:

$$FSR = \frac{\lambda^2}{n_g L} \tag{3}$$

where L is the perimeter of the ring, λ is the wavelength, and n_g is the group index which is defined by Equation 1.

Microring resonators were reported to be used extensively in the field of optical bio-chemical sensing [42, 43]. De Vos et al. demonstrated a label-free biosensor which is based on microring cavities in SOI. The sensing mechanism depends on a resonance wavelength shift amid the change of the surrounding of the cavity. The SOI platform was used because of its high index contrast, which makes it possible to fabricate very small-sized optical cavities resulting in a densely packed chip. Multiarray biosensors for real lab-on-chip devices have also been realized with high sensing capability. The proposed MRR biosensor was functionalized through surface immobilization of the target biomolecule as well as to avoid nonspecific adsorption which can perturb the sensitivity of the biosensor. The bulk refractive index sensitivity was measured to be 70 nm/RIU and the limit of detection (LOD) was 10^{-5} RIU, the biosensor was also capable of detecting concentration of avidin as low as 10 ng/ml [42]. Zecen Zhang et al. demonstrates a mid-infrared biochemical sensor which is based on a microracetrack resonator with a lateral sub-wavelength grating metamaterial cladding on an SOI platform. The sensor was suspended, allowing for a greater modal overlap with the chemical to be tested and as a result boosted the device sensitivity. The dimensions of the suspended device were carefully selected so as to achieve a good compromise between the propagation loss and the sensitivity. The proposed device was expected to have a sensitivity of 337.5 nm/RIU with TE mode, which makes them well-suited for optical bio-chemical sensing applications [43]. The need for much higher sensitivities as well as lower limit of detection has encouraged research to bend towards the development of MRR-based sensors often comprising more than one resonator. In [44] a refractive index based-senor that consist of two cascaded ring resonators and which works based on the Vernier effect was studied, the schematic of the sensor is shown in Figure 2-2. The two rings have different optical roundtrip

lengths and are cascaded in such a way that the drop signal of the first acts as the input of the second. The FSR difference between the two rings is very small resulting in a periodic envelop signal in the transmission spectrum of the cascade. Therefore, a shift in the sensor ring resonance peak, due to a change in the cladding index, will result in a magnified shift in the envelop peak in the spectrum of the cascade. The measured sensitivity was reported to be 2169 nm/RIU with an LOD as low as 8.3×10^{-6} RIU.

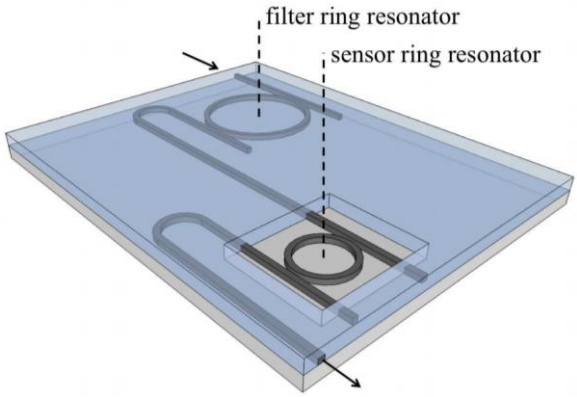


Figure 2-2: Schematic illustration of the two cascaded ring resonators. The devices are buried in a thick cladding layer, with a small opening at the sensor ring to allow for evanescent field-based sensing. The filter ring is shielded from any perturbations [44].

A suspended ring resonator supported by a sub-wavelength grating was proposed by Yousuf et al. [45]. The bottom cladding was etched away in an isotropic manner except near the middle region, leaving a central disk to which the ring was attached to by means of the gratings as depicted in Figure 2-3. A sensitivity of around 503 nm/RIU was estimated for the suspended MRR at 1550 nm for TM mode. Ring resonators were also used in gas sensing applications [46, 47]. Jacob T. Robinson et al. demonstrated a resonant cavity formed by a slotted waveguide. Slot waveguides were used to enhance the light-matter interaction with the gas. The device demonstrated a

sensitivity as high as 490 nm/RIU. Yingying Qiao et al. demonstrated a methane sensor which consists of a long ultra-low loss waveguide and an MRR filter. The sensitivity of the device was based on the change in the optical power with the concentration of the gas via evanescent field absorption at near-infrared band. Two different low-loss waveguides (strip and slot waveguides) with three different materials (silicon, silica, and silicon nitride) were studied for determining the optimal sensing performance at wavelength 1650 nm. Simulation results showed that for a strip waveguide made of silica, and for a waveguide length of 15 cm, the sensitivity can reach up to 8.7 nW/ppm, while for the slot waveguide made of silicon the sensitivity is higher and is estimated to be 13.4 nW/ppm using 15 cm long waveguide.

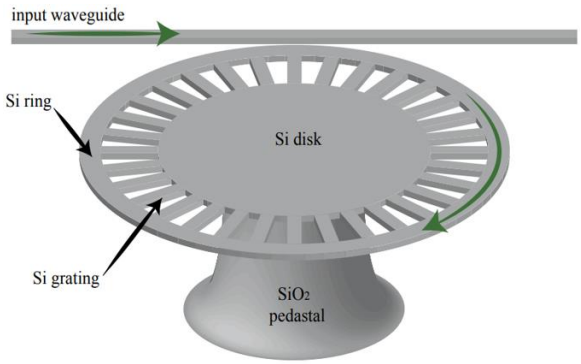


Figure 2-3: Demonstration of the suspended ring resonator, attached to a central disk by sidewall gratings, and supported from the bottom by an oxide pedestal [45].

2.1.2 Bragg Grating Waveguide-based sensors

Segmented SOI waveguides were initially developed by Hochberg et al. [48]. Bragg gratings are another resonant elements which are commonly used in sensing applications because the structure allows for a high field concentration in the cladding which is a fundamental requirement to enhance sensitivity. The structure consists of a Si waveguide with sidewall gratings which are then connected to a silicon body which is optically isolated from the Si waveguide. An illustration of a typical Bragg grating

sensor is shown in Figure 2-4. One of the main characteristic of Bragg grating is the resonance wavelength [49] :

$$\lambda_B = 2n_{eff}\Lambda \tag{4}$$

where, λ_B is the Bragg wavelength, n_{eff} is the effective index, and the grating period is denoted by Λ. Bragg gratings were extensively investigated for use in the field of biosensing [50, 51]. A.S. Jugessur et al. demonstrated a photonic wire Bragg grating device integrated with microfluidic channel for refractive index sensing of fluids or bio-material. Simulation results show that when the cladding refractive index is changed by 0.33 in the presence of water, a shift of about 60 nm is obtained resulting in an average estimated sensitivity of around 5.5×10^{-3} nm^{-1}. P. Prabhathan et al. demonstrated a refractive index based sensor using phase shifted Bragg grating which is appropriate for biosensing applications. The sensing performance was enhanced via careful design to attain certain characteristics (such as narrow band width, and high transmittivity) which are essential to increase the resonant shift with respect to the change in the analyte refractive index. The calculated sensitivity was 90 nm/RIU and the minimum detectable refractive index change was estimated to be 8.1×10^{-5}.

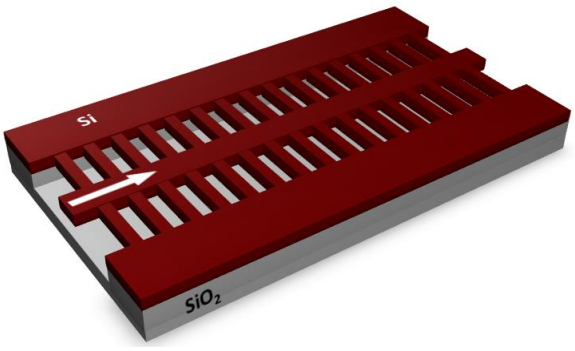


Figure 2-4: Schematic illustration of a suspended Bragg grating waveguide sensor. Suspending the device further increases the modal overlap with the analyte and increases the sensitivity.

The index modulation profile of the Bragg grating can be inscribed in either the core or the cladding region of the waveguide via exposure to ultraviolet light and/or doping [52]. In [53], Tripathi et al. presented a study in which the gratings were inscribed in the GeO_2 doped SiO_2 cladding rather than in the photo insensitive Si core region of the waveguide. The device sensitivity was calculated while operating in TM mode, which has a large field in the sensing region, yielding a predicted sensitivity of 200-740 nm/RIU for a biochemical sample with index varying from 1.33 to 1.63. A biosensor based on phase-shifted Bragg gratings in a slot waveguide on an SOI platform is demonstrated in [54]. A schematic of the sensor is illustrated in Figure 2-5, the sensor was integrated with microfluidic channel for performing a higher biosensing accuracy. The device generates a sharp resonant peak in the measured spectrum by means of introducing a phase shift in the middle of the gratings and exhibits a measured sensitivity of 340 nm/RIU and detection limit of 3×10^{-4} RIU in salt solutions .

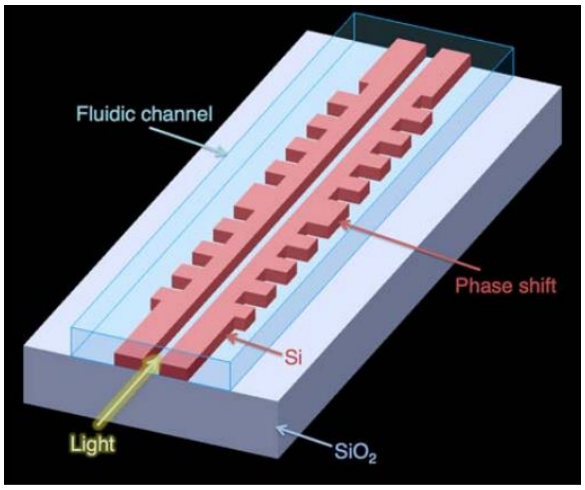


Figure 2-5: Illustration of a phase-shifted Bragg grating in a slot waveguide, on an SOI platform, and integrated with microfluidic channel for enhanced sensing accuracy [54].

A Bragg grating waveguide and a microfluidic channel, co-fabricated within a fused silica glass via femtosecond laser writing and chemical etching was reported in [55]. The sensor works based on evanescent field penetration into the fluid-filled microchannel, with refractive index varying from 1 to 1.452, which thereby induces a wavelength shift. The maximum obtained sensitivity and the minimum detectable index change was 81 nm/RIU and approximately 1.2×10^{-4} RIU, respectively.

2.1.3 Multimode Interferometers-based sensors

Self-imaging is a phenomenon that is realized in multimode interference waveguides (MMI) whereby single or multiple images of the input mode profile are reproduced periodically along the direction of propagation. N×N MMI devices consist of two basic components, the multimode waveguide and the access waveguides which are positioned at symmetrical locations as depicted in Figure 2-6. Multimode interference waveguides are typically wide rectangular waveguides that can support a large number of modes [56]. In general, access waveguides are single mode waveguides, but the use of tapered waveguides at the input and output of the MMI waveguide is sometimes preferred [57]. Tapered access waveguides serve to reduce excess loss, since mode mismatch between the access waveguides and the multimode region is reduced. Tapered waveguides also alleviate crosstalk effect, and are more tolerant to fabrication errors [58, 59].

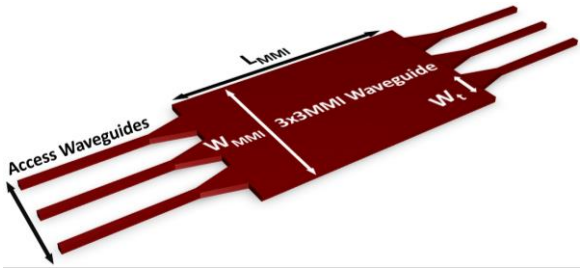


Figure 2-6: Schematic of a 3 X 3 MMI waveguide with a length of L_{MMI} and width of W_{MMI}. The access waveguides are tapered, to reduce excess loss, and have an access width of W_t.

MMIs have been reported to be good candidates to be used as couplers and wavelength demultiplexer devices [60-62]. MMIs have also been widely reported in different sensing applications, including temperature [63], and gas sensing [64]. K.R. Kribich et al. demonstrated a new approach for relative humidity sensing with the aid of MMI couplers. The approach involved the use of the hybrid sol-gel planar technology onto the silicon substrate. The sensor consists of three layers (Figure 2-7): a buffer layer which is used to optically isolate the core from the substrate, a guiding layer and a protective layer. Each layer used the same precursor solutions but in different ratios. The refractive index of the cladding was tuned to have a refractive index difference of 0.01 relative to the core. However, the MMI waveguide region had its top cladding replaced with a sensing coating. The sensing layer material was also prepared via sol-gel precursors, and was designed to have its refractive index increase due to the absorption of water molecules, with values extending from 1.47 at 0 relative humidity to 1.49 at 100% relative humidity. Nonetheless, experimental results showed that the refractive index of the sensing material varied from 1.473 to 1.483 at 40 to 70% relative humidity which fitted within the predicted range. The sensitivity of the device was subsequently measured to be 0.6058 optical intensity arbitrary units/ relative humidity % for a sensor length of 21.2 mm [65].

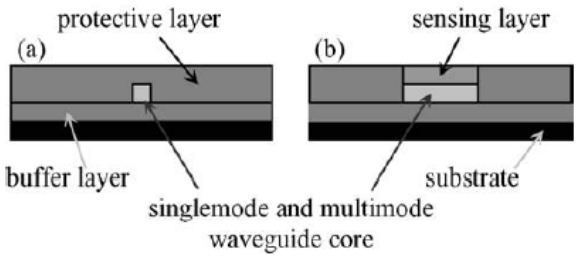


Figure 2-7: Schematic representation of the MMI-based sensor: (a) cross-section of the single mode access waveguide; and (b) cross-section of the multimode sensing region [65].

2.1.4 Mach-Zehnder Interferometer-based sensors

Mach-Zehnder interferometers (MZIs) are optical devices commonly used in sensing applications. The device comprises of an input waveguide which is divided into two parallel arms by means of a y-splitter. The two arm have a physical length which may or may not be equal, depending on the application, and are eventually combined into a single output waveguide by a y-shaped coupler. A schematic representation displaying the basic configuration of an MZI sensor is depicted in Figure 2-8.

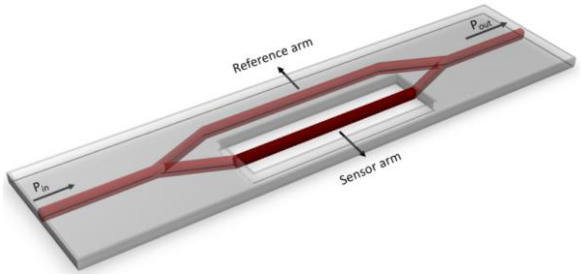


Figure 2-8: Basic configuration of an MZI sensor. The sensing arm is uncovered, allowing the propagating mode to interact with the measurand, whereas the reference arm is wholly protected from undesired effects by being covered with an oxide layer.

An interferometric output response is purely dependent on the phase shift induced between the two light waves interfering at the output of the MZI. The shift could result from a difference in physical length of both arms, or from a different propagation constant in each arm. The aforementioned reasons will result in a difference in optical path lengths. This means that light in each arm will travel at different speeds and then interact constructively or destructively at the y-coupler with varying light intensities due to the optical interference at the output waveguide [35, 66]. The MZI is set to be balanced when the optical path length of each arm is the same (same physical lengths and same effective refractive index in each arm), a condition which is usually irrelevant in applications which require detection of refractive index changes. However, a balanced MZI has a steady response and is more stable to temperature changes [67]. On the other hand, attaining an offset between the optical path lengths of each arm makes the MZI an unbalanced structure, in which the two light waves interfere with one signal being delayed relative to the other resulting in an interferometric output response. The output intensity of the MZI is governed by the following expression (assuming that the loss in each arm is negligible and that the y-splitter divides the input intensity equally to each arm) [68]:

$$I_o = \frac{I_i}{2}[1+\cos\varphi] \tag{5}$$

where I_o is the output intensity, the intensity at the input is denoted by I_i and φ is the phase difference between the two arms of the MZI. The phase difference is given by [68]:

$$\varphi = \beta_{ref} L_{ref} - \beta_{sen} L_{sen} \tag{6}$$

The length of the arms is denoted by L_{ref} and L_{sen}, where $L_{sen} = L_{ref} + \Delta L$ and ΔL is the physical length difference between both arms of the interferometer. The propagation constants are given by:

$$\beta_{ref} = \frac{2\pi n_{eff_ref}}{\lambda_0} \tag{7}$$

$$\beta_{sen} = \frac{2\pi \left(n_{eff_sen} + \Delta n_{eff_sen} \right)}{\lambda_0} \tag{8}$$

where n_{eff_ref} and n_{eff_sen} are the effective refractive indices in the reference and sensor arm, respectively, Δn_{eff_sen} is the induced change in the effective refractive index of the mode in the sensor arm when the refractive index of the cladding is varied, and λ_0 is the operating wavelength.

It is evident in Equation 5 that when the phase difference between the two arms is a multiple of 2π radians, the light beam in each arm will interfere constructively at the output, yielding maximum optical intensity. Likewise, if the phase difference is an odd multiple of π radians the light will interfere destructivity, resulting in a dip in the response of the MZI.

There are two different ways in which the variation in Δn_{eff_sen} can be detected, one way involves operating at a single wavelength and carefully observing the change in MZI output intensity (intensity interrogation method). Another approach is to work in a wavelength range and monitoring the shift in the fringes of the MZI output response, as the cladding medium changes (wavelength interrogation method). In the approach in which the wavelength shift in the spectrum is tracked, the sensitivity of the MZI, with units of nm/RIU, is defined as:

$$S = \frac{\Delta\lambda}{\Delta n_{clad}} \tag{9}$$

where, Δn_{clad} is changes in the cladding refractive index, and $\Delta\lambda$ is the detected wavelength shift. The wavelength shift is effectively dependent on the optical path length difference as expressed below [69]:

$$\Delta\lambda = \frac{\lambda L_{sen} \Delta n_{eff_sen}}{L_{sen} n_{g_sen} - L_{ref} n_{g_ref}} \tag{10}$$

where, n_{g_ref} and n_{g_sen} are the group indices of the propagating mode in the reference and sensing arms, respectively. It is clear from Equation 10 that if the optical path length difference is high, the wavelength shift will be negligible and thus the sensitivity of the MZI sensor, and vice versa. The FSR of the MZI is also dependent on the optical path length difference between the two arms (Equation 11) [70].

$$\Lambda = \frac{\lambda^2}{L_{sen} n_{g_sen} - L_{ref} n_{g_ref}} \tag{11}$$

Similarly, the FSR (the spacing in optical wavelength between two fringes of the MZI response), which has the same denominator as the wavelength shift expression, plays a crucial role in the sensitivity of the MZI. If the optical path length difference is too small, the FSR is too large and the wavelength shift is significant, yielding a high sensitivity. Therefore, in designing an MZI sensor, it is essential to carefully consider the lengths of each arm such that the FSR is wide enough ensuring a greater dynamic range, which is a fundamental requirement to ensure a high sensitivity and avoid wavelength shifts greater than the FSR. In addition, the wavelength shift also depends on the length of the sensor arm as well as the change in the effective refractive index. However, the latter has a lesser impact relative to that imposed by changes in the sensor arm length. Consequently, increasing the sensor arm length, while maintaining

a good balance between the arms, will increase the sensitivity and thereby the sensor resolution (LOD), which is the ability of the MZI sensor to detect the minimum effective refractive index perturbation [38]. The LOD of the optical sensor is given by:

$$LOD = \frac{\Delta\lambda_{\min}}{S} \tag{12}$$

However, it is not convenient to realize very long arms, because it impractically increases the footprint of the device which is a drawback and a challenge in fabrication. A suitable solution, that will make it possible to fabricate mm-long sensors, is to design the MZI arms by exploiting the spiral path configuration which will enhance the function of the sensor without compromising the dimensions of the sensor.

Xianxin Jiang et al. demonstrated biosensor which is based on a cascaded MZI and ring resonator with the Vernier effect via the detection of the wavelength shifts with the change of the refractive index of the cladding. The sensitivity of the cascaded configuration was compared with the MZI alone to illustrate the significance of the Vernier effect in sensing applications. Experimental results indicate that with the Vernier effect, in which the resultant optical transmission is composed of peaks with a periodic envelope function and the total wavelength shift is elevated by a factor, the sensitivity can reach up to 21,500 nm/RIU in contrast to the 2870 nm/RIU of the MZI alone. The sensitivity values were obtained with careful selection of the relation between the sensing and the reference arms, and using an aqueous solution of NaCl with varying concentrations in increments of 0.05%. A separate biosensing experiment was then conducted, to examine the interaction between goat and antigoat immunoglobulin G (IgG) pairs, via surface functionalization and the use of a

Polydimethylsiloxane (PDMS) microchannel. Results demonstrated the ability of the cascaded sensor to achieve a wavelength shift of around 0.5 nm with only 1ng/ml IgG relative to the 0.035 nm achieved by the MZI alone, thus, making the cascaded sensor a favorable candidate in sensing applications [69]. Raghi S. El Shamy et al. designed and simulated two configurations of an MZI sensor for gas sensing in the mid-infrared region of the spectrum. Mid-infrared wavelength range was given preference to due to the intrinsic gas selectivity as every gas has its unique absorption line in that range. Wavelength of 5 μm was selected, particularly where the absorption peak of carbon monoxide occurs. Both wavelength and intensity interrogation methods were studied, where measuring the absorption of light as a consequence of its interaction with gases has proven to have a lower LOD relative to the wavelength shift detection approach. The two designs were based on whether a zero or low sensitivity was to be realized on the reference arm. In the case of low sensitivity, careful selection of the reference arm width was to be taken into account to minimize the interaction of the mode with the analyte, for this configuration both arms of the sensor are suspended. However, for the case of zero sensitivity, the senor was covered with oxide (Sapphire), which has a low absorption at the operating wavelength, with an opening only at the sensing region. For the simulation, TE mode was selected, since in this mode the sensitivity of the reference arm decreases with increasing the width as compared with the TM mode in which the thickness of the waveguide needs to be varied. The latter configuration was used in the wavelength interrogation method, in which the device sensitivity can reach up to 7028 nm/RIU with an intensity LOD around 5.6×10^{-3} RIU, and a wavelength LOD around 5.5×10^{-3} RIU with careful selection of sensor arm length, width, and y coupler dimensions. On the other hand, the suspended MZI design demonstrated an intensity LOD as low as 2.7×10^{-3} RIU [71]. In [72], an SOI MZI senor with a sub-

wavelength grating (SWG) sensor arm is studied for lab-on-chip applications. The sensor was deliberately designed so as to ensure almost flat phase difference between the guided light in each arm of the sensor, for a desired wavelength range (around 1550 nm). The sensor arm was designed with SWG taper on either side of the SWG arm in order to reduce mismatch loses between the single mode waveguide and the SWG waveguide. The reference arm was designed to have the same length and width as the sensor arm. The width was chosen to be too wide so as to reduce the sensitivity of the reference arm to changes in the cladding, an approach which avoids an additional fabrication step such as the deposition of oxide cladding to accomplish isolation. In addition, the device length of the SWG arm was carefully selected to ensure that for a given set of cladding refractive indices, the output response is adjusted to the region where the transmission is linear and the interferometric pattern exhibits a wavelength independency. The 3D schematic illustration of the SOI sensor is depicted in Figure 2-9, a 50:50 MMI device was used instead of a y-splitter/combiner. Simulation results has demonstrated a decrease in the phase difference between the two arms with an increase in the cladding refractive index. Experimentally, the phase sensitivity was measured to be 931 rad/RIU/mm with a device length of 185 μm and an increase in the cladding refractive index by 0.01.

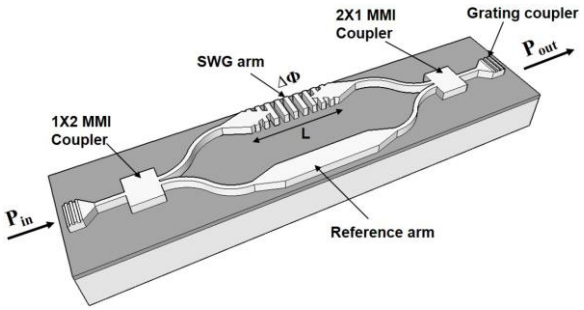


Figure 2-9: Illustration of an MZI sensor with an SWG sensor arm, and a reference arm adjusted to exhibit no sensitivity to changes in cladding refractive index [72].

Owen Marsh et al. demonstrated a fluid sensor which is comprised of a ring-assisted MZI (RAMZI) fabricated with slot waveguides. The sensor was realized using two different optical elements for coupling: directional coupler (DC) and MMI coupler, and the results were subsequently compared with the performance of a standard MRR sensor with both coupling elements. Different sucrose concentrations varying from 1% to 6% with increments of 1%, which translates into a linear change of the refractive index from 1.333 to 1.3418, was used for testing sensor performance. Experimental results show that for RAMZI with DC, the measured sensitivity is 203.3±15.3 nm/RIU versus the MRR with DC which exhibited a sensitivity around 299.7±17.4 nm/RIU, however, the RAMZI-DC has a much larger FSR and thus provides a larger range for measurement. On the other hand, for the RAMZI with MMI the measured sensitivity was 177.2±12.0 nm/RIU which was approximately 2.7 times larger than the sensitivity of the MRR with MMI sensor, and also demonstrated a greater dynamic range for measurement [73]. Another MZI sensor which was based on a silicon nitride slot waveguide was demonstrated for biosensing applications in [74]. The sensor arm was made of a slot waveguide which is known to enhance light-analyte interaction, and thus increase the sensitivity, while the reference arm consisted of a strip waveguide (buried waveguide). A mode converter was used in the sensing arm at the transition between the strip and slot waveguides as shown in Figure 2-10. The entire device was buried in an oxide cladding expect for a small window at the slot waveguide region for the light-analyte interaction to take place. Different concentrations of NaCl solution were used to quantity the bulk sensitivity of the device which was measured to be 1864 π/RIU with a slot waveguide length of 7 mm and the LOD was as low as 5.4×10^{-6} RIU. The biosensing experiment was then conducted to measure the biotin-

streptavidin interaction via surface functionalization for which a phase change of 0.14π was obtained for a 1 pg/ml concentration of streptavidin solution.

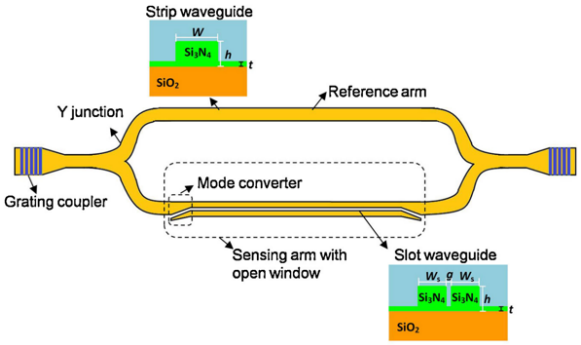


Figure 2-10: Illustration of an MZI biosensor comprised of a silicon nitride strip reference waveguide and a silicon nitride slot sensor waveguide [74].

2.2 Summary

The chapter provides a meticulous review on the fundamental optical components and their role in optical sensing applications. The optical components summarized in this chapter include Microring Resonator-based sensors, Bragg Grating Waveguide-based sensors, Multimode Interferometers-based sensors, and Mach-Zehnder Interferometer-based sensors. Sensing applications range from a real-time detection of induced cladding refractive index change to label-free, selective detection of various biological analytes. The focus is to determine the optical sensor which performs best with regards to high sensitivity, very low limit of detection, compactness and uncomplicated fabrication procedures. Highest sensitivities were observed in devices in which the Vernier effect was employed. Vernier effect-based sensors also exhibited a low limit of detection which makes them suitable candidates in optical sensing technology. However, one limitation is the need for a large wavelength range so as to be able to track the envelope curve. Mach-Zehnder Interferometer-based sensors have the advantage of low cost, ease of fabrication and measuring, and high refractive index

sensitivity. However, a significant drawback in MZI-based sensors is their susceptibility to the environmental temperature. Consequently, the current work pays particular attention to simple potential approach in tackling this problem for maximizing the capability of MZI devices in optical sensing.

Chapter 3: Simulation Tools

In this chapter, the simulation tools which were used in the modelling of the proposed devices will be described briefly. Different simulation tools were used to observe the mode profiles along the cross-section of the device and study how the optical mode properties vary along the direction of propagation on account of reflection, scattering or interference.

3.1 Finite Difference Time-Domain FDTD

Three-dimensional (3D) Maxwell equations can be solved using a numerical method such as the FDTD technique which employs spatial and temporal discretization methods on a multi-dimensional grid. It is a technique in which it is possible to analyze the interaction of light with complex structures with features that are at a sub-wavelength-scale. Accurate calculation of the Maxwell equations can be reached by reducing the spatial discretization of the volume i.e. by using smaller mesh size [75]. The method is based on volumetric sampling of the electric field and magnetic field staggered distribution within a specific grid over a period of time, which is an approach that was initially proposed by Yee [76]. The nature of Maxwell equations imply that a change in the electric field in time is reliant on the spatial variation of the magnetic field (the curl) and vice versa. This results in basic FDTD time-stepping relation whereby, at any point in space, the present value of the electric/magnetic field in time is reliant on the previous value of the electric/magnetic field and the curl of the local distribution of magnetic/electric field in space. The electric and magnetic field components are positioned about a unit cell of a computational grid as shown in Figure 3-1, where electric and magnetic field components are calculated at half-time steps. In this work, Lumerical FDTD solutions was used [77].

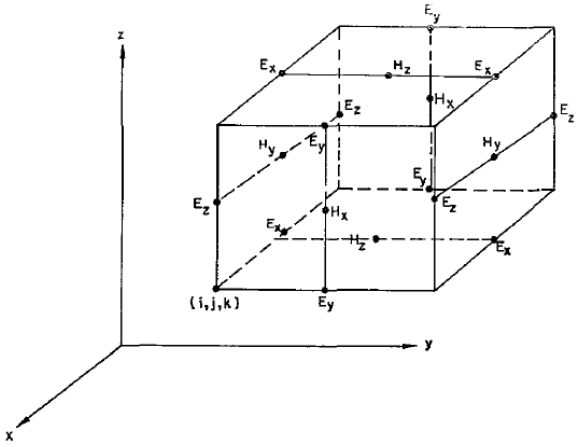


Figure 3-1: Position of the electric and magnetic field components on the unit Yee cell [76].

A drawback of FDTD is that a computational domain must be established, which requires a long computational time. The software permits desired structures to be defined within the simulation region, subsequently the material of every cell within the computational domain must be identified. Once the aforementioned properties are established, a source is selected which in this project is the mode source, which is used to launch a fundamental mode of the specified polarization. A boundary condition must be defined, Perfect Matching Layers (PML) is the most common and is used in this work. PML absorbs all the light exiting the simulation region and eliminates the reflection at the boundaries. It can also reduce computational time if a symmetry is defined at the z-direction.

3.2 Mode Solver-MODE

Mode solver (or Eigenmode solver) is a simulation tool in which the optical modes in a cross-section of a waveguide at a specific wavelength is computed. The mode solving software used in this project is Lumerical MODE Solutions [78], which uses a finite-difference eigenmode (FDE) algorithm based on a solver presented by Zhu and Brown [79]. Similar to FDTD simulations, waveguide structure, grid material,

boundary condition, and mesh accuracy must be defined. The simulation can be conducted for a single frequency or a range of frequencies which is essential to study waveguide dispersion. Once the meshing of the desired structure is completed, the mode profiles, effective indices, and the losses are then computed by solving Maxwell's equations on a cross-sectional mesh of the waveguide [75].

3.3 Photonic Integrated Circuit Simulator – INTERCONNECT

A photonic integrated circuit simulator is a simulation tool which is fast, stable, and accurate for analyzing, and modelling the design of integrated optical circuits. Device models which are constructed with Lumerical's FDTD and MODE Solutions simulation tools can be easily incorporated within INTERCONNECT circuits, allowing for the inclusion of complex device-level physics to simulate realistic photonic integrated circuits, but with a significantly faster computational time compared to the 3D FDTD simulations. In this work Lumerical INTERCONNECT [80] is used.

3.4 S- Parameters

The S-Parameter simulator (SPS) solver [81] is a simulator that serves in calculating the general response of a device in frequency domain. The S-parameters can be extracted from FDTD simulations and they comprise both: amplitude and phase responses. A standard sign and phase convention of the electric and magnetic fields of the forward and backward modes should be used for all ports of a component. S-parameters can be imported into INTERCONNECT for conducting system-level simulations; however, the imported data has to comply with the tests for reciprocity as well as passivity to ensure reliability with regards to accuracy and convergence [82].

3.5 Summary

In this work, Lumerical tools were extensively used in order to design the proposed structures with optimal functionality and enhanced sensitivities. Mode Solver helped determine the mode profile and acquiring the essential mode properties, which are necessary for the design of the proposed structures. S-Parameters, imported into INTERCONNECT, were used in structures which are very large and thus, require long computational time to be simulated via 3D-FDTD simulation tool.

Chapter 4: Device Design and Modelling

Optical sensors based on Microring resonators, Bragg gratings, Multimode interferometers, and Mach-Zehnder interferometers have been discussed extensively in Integrated Optical Sensors chapter. Different waveguide configurations as well as various approaches to make the sensitivity as high as possible were demonstrated. The current chapter draws special attention to the proposed MZI structures and their application in optical sensing. Different simulation tools were used to design the MZI sensors with optimum performance at the desired wavelength and to ensure single mode operation for the different waveguide configurations. All proposed optical sensor structures were built on an SOI platform and were designed using KLayout which is a software that serves in generating layouts in GDSII file format.

4.1 Partially Exposed MZI

In this work, the sensing mechanism is based on evanescent field penetration into the cladding medium, ultimately resulting in a wavelength shift which is a critical figure of merit to quantify the sensor performance. Two sensor configurations are studied in this work, partially exposed and suspended MZI sensors, where the latter will be discussed in the following section. The schematic representation of the partially exposed MZI sensor arms is shown in Figure 4-1. The reference arm is an SOI buried waveguide and thus the guided light is isolated from the target analyte. However, the sensor arm is an SOI rib waveguide with a 2 µm thick top oxide cover and a window in a section of the waveguide of length ~10827 µm and a depth through the thick oxide to the Si slab.

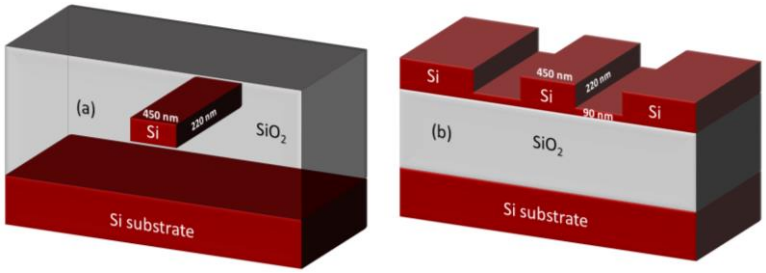


Figure 4-1: Schematic representation of the partially exposed MZI sensor arms: (a) cross-section of the reference arm; and (b) cross-section of the sensor arm.

The reference waveguide has a width of 450 nm and a height of 220 nm, the typical dimension for a single mode operation with insignificant propagation losses attributable to side wall roughness. The sensing arm has a width of 450 nm and a height of 220 nm, patterned on a 90 nm-thick Si slab, as depicted in Figure 4-1. The sensor is designed to operate at 1550 nm, where the refractive index of SiO_2 and Si is 1.444 and 3.476, respectively for TE mode. The length of each arm is around 1 cm long, necessitating the spiral configuration of the arms for a more compact device and a better sensor performance. The arms were designed to be very long to facilitate the detection of very small fluctuations in the claddings refractive index, in presence of the measurand, and thus boosting the sensitivity of the device [15, 83]. Each arm has a different waveguide configuration and are a little off in terms of physical length, making the MZI sensor slightly unbalanced, which is a vital requirement in sensing applications. Unbalanced MZI sensor with a low optical path length difference yields an interferometric response with an FSR wide enough, ensuring that the induced wavelength shift is not larger than the period, for an accurate sensitivity measurement. Another important parameter is the contrast or the fringe visibility of the MZI response which is determined by the proportionality at the y-splitter. Even splitting and low

propagation loss in each arm results in a high visibility. The layout of the y-branch used in each side of the partially exposed MZI sensor is the same as in [68].

A common and a considerable challenge with MZI sensors is the influence of the temperature change on the sensor performance which results in uncertainty in the measured sensitivity. The effect of the temperature change is a consequence of the high thermo-optic (TO) coefficient of silicon (~1.86×10^{-4} K^{-1}) [84]. Numerous researches has been conducted to establish efficient ways to eliminate the effect of the undesired temperature drift on optical sensors. Some approaches are not CMOS process-compatible, this includes the use of materials which take away the effect of the high TO coefficient of Si, and others are power hungry [85]. In [86-89], a polymer overlay cladding with a negative TO coefficient was used, and in contrast with Si, an increase in temperature reduces the refractive index of these materials, thereby counteracting the effect of the temperature drift on the function of the sensor. Engineering the optical mode confinement in Titania (TiO_2) cladding, which has a high negative TO coefficient, is another attractive CMOS process-compatible approach in mitigating the TO effect on silicon based optical sensors. This approach requires careful design of the waveguide geometry and optimal cladding thickness for effective temperature compensation over a wide range of operating temperatures [90, 91]. Diligent selection of the arm dimensions is another intriguing approach to tackle the temperature-dependent wavelength shift problem in optical sensors [70, 85, 92]. Another approach involves setting a different polarization in each arm of the MZI device [93], or using different waveguide material composition with the aim of balancing the thermo-optic effect in each arm [94]. Most solutions involve obtaining a difference in the MZI arms (varying the configuration, polarization etc.) such that an opposing temperature-dependent phase change is induced in each arm, thereby

reducing the temperature-dependent wavelength shift of the MZI sensor. In this work, the temperature effect is compensated for with the aid of a ring resonator which is positioned at the end of the MZI sensor, in a way that the output waveguide of the Y-combiner serves as the through port of the ring resonator. In this regard, the measured output response subsumes the interferometric response of the standard MZI sensor as well as the through port response of the ring resonator. Figure 4-2 shows the GDS layout of the partially exposed MZI sensor, with the temperature-compensating ring resonator placed at the vicinity of the output waveguide. The temperature compensating MRR has the following parameters: 220 nm silicon layer thickness, a ring core width of 450 nm and a radius dimension 10 μm. The coupling gap between the output waveguide and the MRR was set to be 300 nm, and using Equation 3, the FSR was calculated to be around 9 nm for TE mode, at 1550 nm.

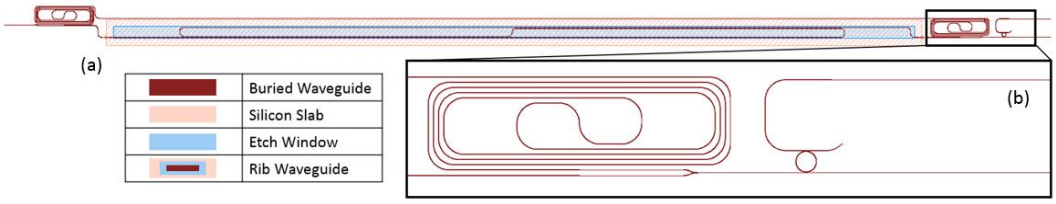


Figure 4-2: Illustration of the partially exposed 1-cm long MZI sensor on a 90 nm Si slab: (a) GDS layout of the MZI sensor; and (b) zoom-in-view of the temperature-compensating MRR.

The MRR is a rectangular waveguide buried in 2 μm thick SiO_2 to prevent exposing the propagating mode to any undesired perturbations, and so any observed shift in the response is solely a result of a temperature gradient within the waveguide in presence of the measurand. The resonance wavelength shift of the temperature-compensating MRR is expressed by Equation 13.

$$\Delta\lambda = \frac{\lambda \Delta n_{eff_sen}}{n_{eff_sen}} \tag{13}$$

where, $\Delta\lambda$ is the measured resonance wavelength shift, λ is the operating wavelength, and Δn_{eff_sen} is the determined effective index shift caused by the change in temperature. Equation 13 is exploited in order to determine the temperature change induced during the experiment. The n_{eff_sen} is calculated using MODE solver, and along with the detected temperature-dependent wavelength shift ($\Delta\lambda$), the Δn_{eff_sen}, at the wavelength of interest (λ), can then be deduced. Subsequently, several MODE solver simulations are carried out to quantify the corresponding change in the refractive index of Si which results in the calculated Δn_{eff_sen}. Using the TO coefficient of Si (TOC_{Si}) in company with the change in the refractive index of Si (Δn_{Si}), the temperature drift (ΔT) resulted during the experiment can then be measured using the following equation:

$$\Delta T = \frac{\Delta n_{Si}}{TOC_{Si}} \tag{14}$$

Determining the temperature drift would assist in eliminating the temperature impact on the MZI sensor and thus avoid ambiguity in the sensor performance. This is an accurate approach in compensating for the temperature-dependent wavelength shift in the MZI sensor. The accuracy is explained by the fact that the wavelength shift in both: MZI due to gases and temperature drift, as well as MRR due to temperature alone are tracked concurrently.

The functionality of the sensor is studied via MODE solver simulations and Matlab script. The medium cladding refractive index, in the opening at the sensor arm, was varied from the refractive index of air, which is approximately 1.00027 at 1550 nm [95], to 1.00127 and the corresponding wavelength shift (Equation 10) was then

calculated for every increment to estimate the sensitivity of the device. The sensitivity of the partially exposed MZI sensor, which is defined by the slope of the plot observed in Figure 4-3, was estimated to be about 2071 nm/RIU.

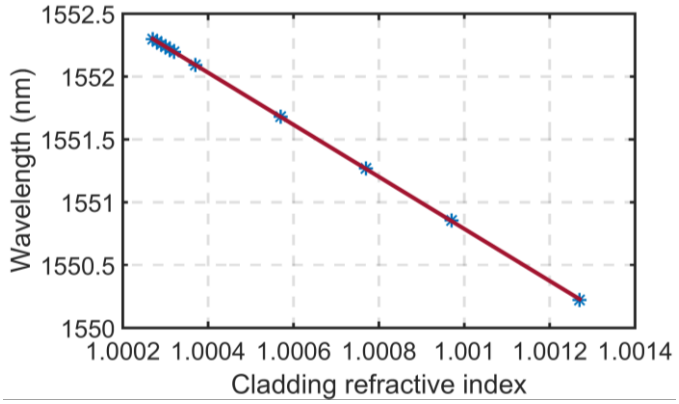


Figure 4-3: Calculated wavelength shift for the partially exposed MZI sensor with different cladding refractive index.

4.2 Suspended MZI

Another structure is proposed in which the sensor arm of the MZI is suspended (silicon oxide cladding is removed from the top and the bottom) via the use of the segmented wave guiding technique. The reference arm is a buried waveguide with a width of 450 nm and a height of 220 nm. The sensor arm is a 450 nm wide and 220 nm thick suspended rectangular waveguide with sidewall gratings. The gratings act as the optical elements which bridge the suspended waveguide to a Si block, which is not released, for mechanical support as shown in Figure 2-4. The period of the gratings is 400 nm (200 nm gap and 200 nm Si support). The sensor configuration performs best with TM-polarized light, given that TM mode eases the fabrication constraints of the gratings with period that satisfy the first-order Bragg condition, owing to its smaller effective refractive index relative to TE mode. The dimensions were carefully chosen such that, TM mode propagates with negligible propagation loss while enabling a

further delocalized profile for achieving higher sensitivities. The length of the reference arm is set to be 750 μm while the sensor arm is set to be 850 μm long, hence, the suspended MZI is unbalanced with regard to the optical path length of each arm. In common with the partially exposed MZI, the suspended MZI is also prone to perturbations due to the temperature drift. A ring resonator is also used here to counterbalance the temperature influence. Figure 4-4 shows the GDS layout of the suspended MZI sensor, with the temperature-compensating ring resonator placed at the vicinity of the output waveguide. The temperature compensating MRR has the following parameters: 220 nm silicon layer thickness, a ring core width of 450 nm and a radius dimension 20 μm. The coupling gap between the output waveguide and the MRR was set to be 500 nm, and using Equation 3, the FSR was calculated to be around 5 nm for TM mode, at 1550 nm.

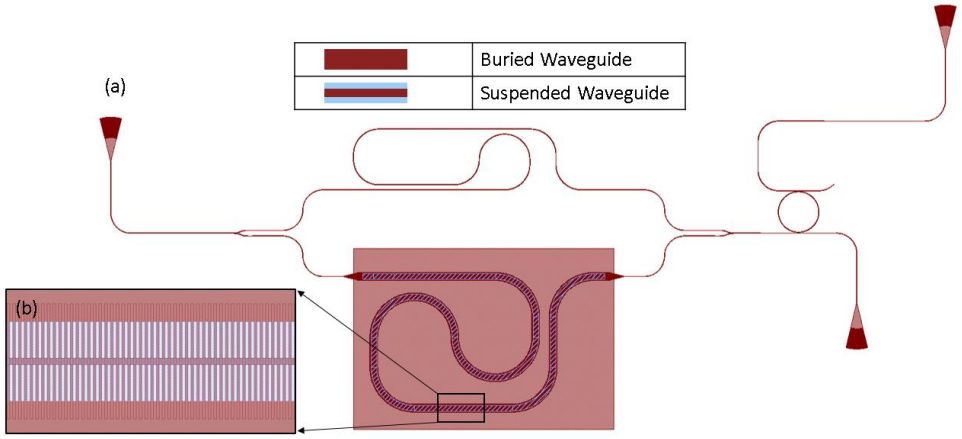


Figure 4-4: Illustration of the suspended MZI sensor using sidewall gratings: (a) GDS layout of the MZI sensor; and (b) zoom-in-view of the suspended sensor arm.

The simulation of the sensor was conducted via the use of ports in Lumerical FDTD along with Optical N-Port S-parameter (SPAR) element in INTERCONNECT. The length and the complexity of the suspended sensor, causes the otherwise numerically rigorous and accurate 3D FDTD simulations, to be computationally intensive and

more time consuming. Therefore, in order to ensure numerical convergence and effectively lessen the numerical errors and simulation time duration, an alternate approach was preselected. The ports in Lumerical FDTD serve as both, the mode source as well as the field monitor, and when used together with the S-parameter sweep tool, generates the S-parameters. The S-parameters, as a function of wavelength, are then imported into INTERCONNECT, in which a compact model is used, so as to study the behavior and performance of the sensor. Figure 4-5 depicts a screenshot of the suspended arm from a 3D Lumerical FDTD simulation which was conducted for only a length of 50 μm instead of simulating the actual total length, which is 850 μm, to reduce the simulation time. The ports were carefully positioned such that a total length of 50 μm, from the beginning of high to the end of low refractive index regions of the periodic grating structures, is attained. In this manner, the extracted S-parameters can be imported to a series of SPAR elements in INTERCONNECT, to complete the actual total length of the sensor arm, with higher simulation accuracy and significantly faster computation time. For the reference arm, MODE waveguide in INTERCONNECT is used in which wavelength dependent properties, among which are the effective and the group indices of the buried waveguide, can be directly imported from MODE Solutions, resulting in accurate simulation results. Figure 4-6 shows a screenshot of the simplified sensor structure in INERCONNECT. The characteristic of the sensor was tested by an Optical network analyzer (ONA) element which is used for both, injecting light into the MZI and determining the response of the sensor as a function of wavelength.

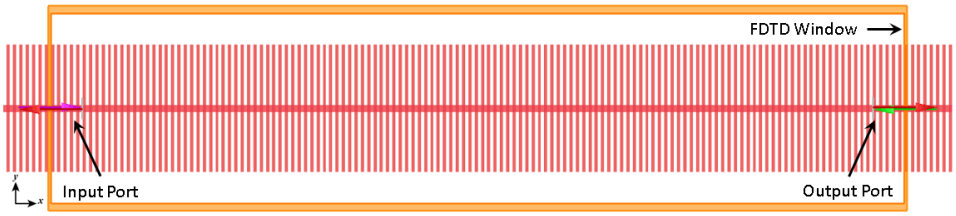


Figure 4-5: Schematic of the suspended sensor arm simulated with a length of only 50 µm using two ports and the S-parameter sweep tool in Lumerical FDTD.

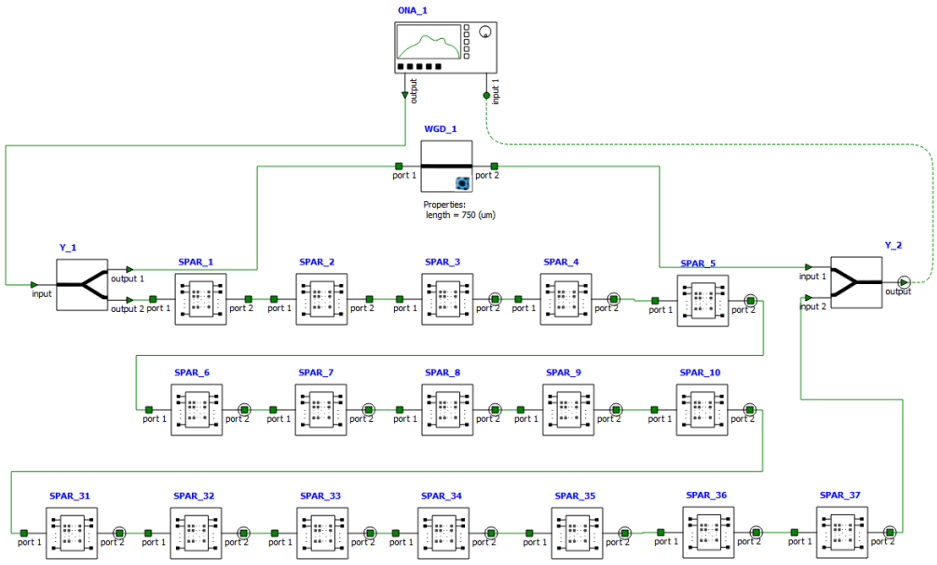


Figure 4-6: Illustration of the designed suspended MZI sensor model in Lumerical INTERCONNECT.

The 3D FDTD simulation was conducted in different gaseous medium: He, and N_2, with the aim of computing the bulk sensitivity of the device. The simulation in INTERCONNECT was performed for both gases, the results of which are shown in Figure 4-7.

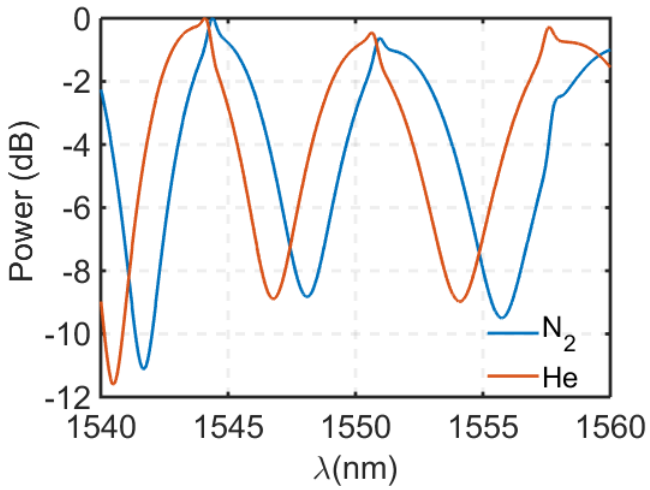


Figure 4-7: Simulated gas sensitivity for the suspended MZI.

INTERCONNECT simulation results demonstrated that the response shifts towards a shorter wavelength when the cladding medium alters from N_2 to He gas. Sensitivity as high as 5500 nm/RIU, near 1550 nm wavelength, was observed, proving the potential of the suspended structure in sensing applications.

4.3 MZI Bio-sensors

4.3.1 Device Sample Preparation and Sensor Integration

The MZI sensors are to be applied in a biosensing application, a project that was inspired to aid in COVID19 Microsystem-based detection. In any manner, SOI-based optical sensors cannot in their essence, directly recognize the presence of a certain biological molecule and differentiate them from other particles in the surrounding environment. This feature can be ascribed to optical sensors via an additional functional layer which facilitates the bonding of specific receptors onto the target molecules to be identified. Consequently, the surface of the optical sensor presented in this work needs to be altered through chemical functionalization with certain receptors for the desired biological targets to be translated into a measurable signal. In

this endeavor, label-free quantification is favored for a direct, real-time detection of the targets captured by immobilized receptors on the surface of the optical sensor.

A great virtue of label-free optical based biosensors is their capability to perceive biological sample including proteins and DNA. This characteristic of optical biosensors is attributed to the fact that the permittivity (ϵ) of biological samples dominates that of air and water. Accordingly, the target bio-samples can potentially reduce the propagation velocity of electromagnetic waves that get through them, interpreting this change into a measurable quantity which is proportional to the magnitude of the samples residing on the surface of the sensor [96]. In this regard, distinguishing bio-samples is accomplished owing to the coating of the functional layer whilst measurement of the quantity of bio-samples which is realized by a detected change in the sensor's optical characteristic.

Therefore, the objective is to incorporate integrated chips with a simple sensing protocol to execute a Lab-on-Chip sensing performances in order to reduce measurement complexity and offer convenient and easy-to-use medical diagnostic apparatus. Typically, in favor of comprehending biological samples, technologies that enable the detection and amplification of small bio-samples are opted for. Polymerase chain reaction (PCR) is the most popular and commonly used in-vitro technique for amplifying a certain segment/target of the DNA, producing up to millions of copies of the target region for analysis [97]. However, despite the fast detection speed of the PCR technique, they are expensive and on that account an alternate label-free technique is to be applied to the work presented in this thesis, to detect biological samples from a complete cell to a single molecule with exceptional abilities.

4.3.2 Cell Lysis Technique

The technique starts with the extraction and purification of cell contents through a process referred to as cell lysis. Cell lysis is a way to rupture the cell membrane so as to free the intracellular substances such as DNA, RNA, or proteins. Cell lysis based on biological, chemical or physical methods are commonly practiced [98]. However, in this work the focus will be on physical-based cell lysis and particularly acoustic lysis which expedites the cell membrane rupture through ultrasonic cavitation (formation of bubbles). A Piezoelectric Micromachined Ultrasonic Transducers (PMUTs) consolidated with microfluidic devices will be used for agitating and disrupting cell membranes with high stability and cell lysis efficiency. A cross-section of the PMUT designed for this application is shown in Figure 4-8, an array of eight PMUTs was designed, as shown in the GDS in Figure 4-9, for enhanced performance. Accordingly, a sufficient gap between each PMUT cell, connected in parallel in the array, was taken into account to reduce acoustic interactions between neighboring cells that lead to a crosstalk effect which can degrade the performance of the transducer.

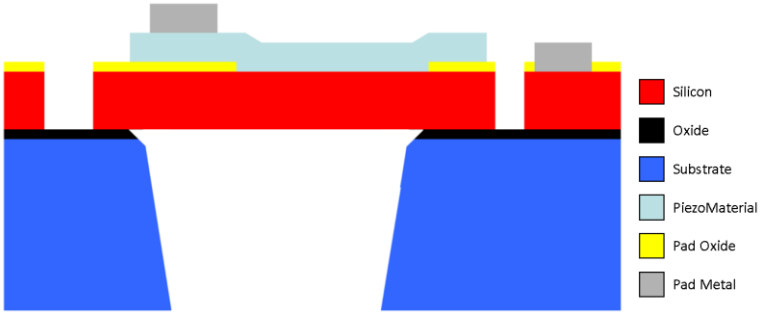


Figure 4-8: Cross-sectional view showing all layers of the PMUT device.

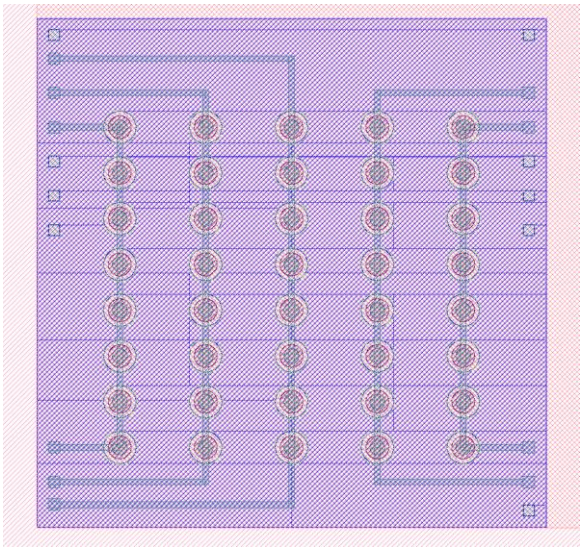


Figure 4-9: GDs layout including an array of eight PMUT cells.

The PMUT device consist of a stack of layers of different materials on an SOI platform. The stack consists of an aluminum nitride layer, the piezoelectric layer, a metal stack consisting of 20 nm chrome and 1000 nm aluminum, a silicon layer and a SiO_2 layer. The device had a backside cavity, with a diameter of 200 μm, which was necessary to release the structure and allow the circular membrane to freely vibrate. All PMUT cells in the array had a top electrode with a diameter equal to 65% of the cavity diameter. The design dimensions were chosen to achieve a resonant frequency of approximately 3 MHz which is a frequency that enables cell lysis with a practical, easy-to-fabricate membrane size. Increasing the frequency boosts cell lysis efficiency, however it would mean decreasing the size of the membrane. The downside of decreasing the membrane size is the reduced area of interaction as well as the challenges encountered in fabrication. Therefore, for this work the minimum membrane size used is limited by the capabilities of the foundry which in this case is 200 μm in diameter for which 3 MHz is the resonant frequency.

A preliminarily simulations were conducted to study the behavior of a single PMUT cell using COMSOL Multiphysics 5.4. The purpose of the simulation was to study the behavior of high frequency PMUTs in air and water with the layout dimensions and to test if they operate in the frequency range of interest (around 3 MHz). Another objective is to study the ultrasonic interaction with fluids in microfluidic channels, including the influence level of pressure and stress. Figure 4-10 shows the COMSOL model of a single PMUT cell connected to a Si-based microfluidic channel, the cavity and the channel are air-filled or water-filled, respectively. The PMUT device had an air-dome domain at the top with a non-reflecting boundary condition to be able to absorb the radiating wave.

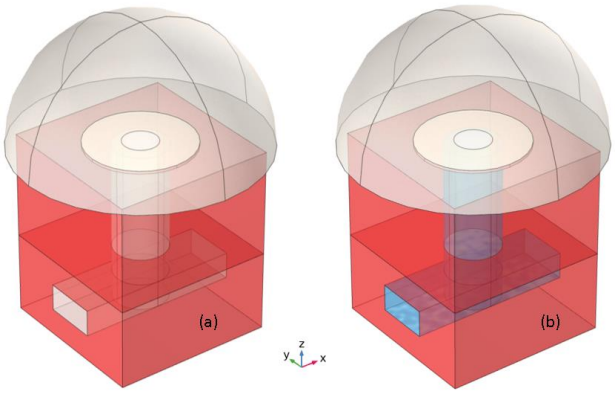


Figure 4-10: COMSOL model of proposed PMUT design interacting with: (a) air; and (b) water in the microfluidic channel.

Thus far, simulation only for air-filled cavity was conducted. A 5 V-peak pulse was applied on the PMUTs' top electrode, while the bottom electrode was grounded. The steady-state frequency response was obtained after applying a 5 V-peak across the electrodes and performing a frequency sweep around the resonant frequency which was estimated to be around 3.7815 MHz, which is close to the desired frequency. Figure 4-11 demonstrates the obtained displacement as well as the admittance

spectrum at the center of the membrane for a frequency range of 3.65-3.95 MHz in an air-filled cavity.

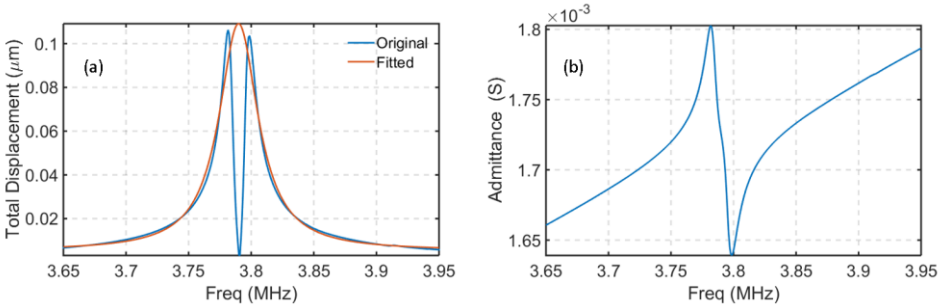


Figure 4-11: (a) Simulated displacement frequency response; and (b) simulated admittance spectrum for a 200 μm diameter PMUT in air.

Acoustic pressure simulation results of a 200 μm diameter PMUT in air-filled cavity at resonance and off-resonance are shown in Figure 4-12 (a) and (b), respectively, where the maximum acoustic pressure amplitude of 8890 Pa was obtained in air at resonance. The acoustic pressure is proportional to the displacement of the membrane and is generated by areas of compression and rarefaction in the medium in the cavity.

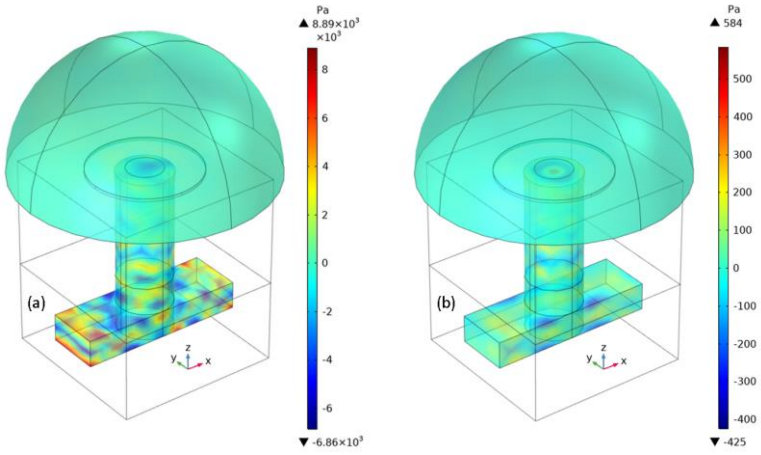


Figure 4-12: Acoustic pressure simulation results: (a) at resonance; and (b) off-resonance in a 200 μm diameter PMUT with air-filled cavity.

Following the complete study on how the PMUT behaves in air medium with regards to acoustic pressure, a model in which a layer of protein of 5 nm was placed in the cavity and simulated, results of which are shown in Figure 4-13 at resonance, which was found to be 3.195 MHz. Figure 4-13 (a) shows the acoustic pressure simulation while Figure 4-13 (b) shows the sound level simulation results, where it is clearly evident that in the latter most of the power was reflected back due to the large acoustic impedance mismatch at the protein-air interface. Therefore, no acoustic pressure was transferred to the air medium since the boundary between the two mediums (protein and air) contained most of the acoustic energy. This indicates that a cleaning of the channel after conducting an experiment might be necessary in order to remove any adsorbed proteins and avoid cross-contamination.

These studies have confirmed that the PMUT, with the aforementioned dimensions, have a resonant frequency which is well-matched with the expected. A brief overview on how the acoustic pressure varies in air-filled cavity and channel was also provided in the preliminary analysis conducted with COMSOL.

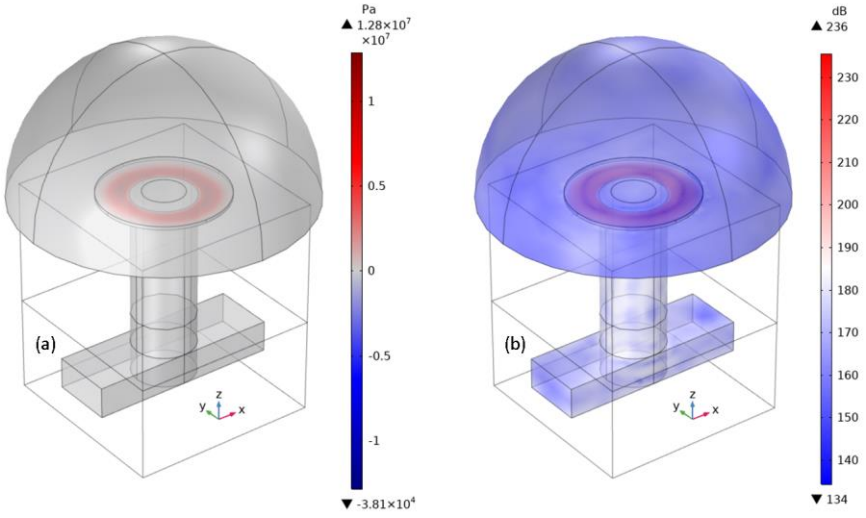


Figure 4-13: Simulation results showing: (a) acoustic pressure; and (b) the sound level when a 5nm layer of protein was added on the 200 μm PMUT device.

4.4 AWG-MMI Integrated Spectrometer

In the previous chapters, we emphasized on different optical on-chip sensors which are mainly established via the evanescent field spreading into the cladding medium. Some sensors require a complicated systematic study like surface functionalization in order to boost their sensitivity and selectivity to analytes. However, optical sensors are in most cases simple and offer superior performances in sensing applications. The transduction of the target molecules being measured into optical/electrical signals, which are readily quantified by a computer, is usually done via an external readout system. Expensive off-chip bulky systems like in the case of a tunable laser with photodetector or an optical spectrum analyzer with a broadband source, have urged the need for photonic chip sensors with integrated spectroscopic systems [99]. Waveguide integrated spectrometers comprise an arrayed waveguide grating (AWG) which generally consist of two free propagation regions (FPR) connected via an array of waveguides with varying lengths [100-102].

In this work, a miniaturized spectrometer is fulfilled by means of a wavelength demultiplexer (DeMux) circuit. The DeMux is composed of AWGs with MMIs replacing the conventional FPR regions. Wavelength-division (de)multiplexing (WDM) is recognized as the key technology for maximizing transmission capacity and flexibility of optical communication systems [103, 104]. Alternate established multiplexing techniques include spatial-division multiplexing (SDM) in which multiple channels of message signals are conveyed via different modes or cores [105, 106], and mode division multiplexing (MDM) [107-109]. Several configurations of the WDM have been implemented, some of which are diffraction based and some are interference based. For instance diffraction based WDM comprise dispersive optical components such as diffraction gratings in which the input signal is diffracted

backward and focused into a succession of output waveguides that are spaced as per the desired channel spacing [110, 111]. Interference based multiplexers are reliant on optical filters which leverage interferometric devices in order to facilitate wavelength selectivity. In [112] optical filters based on a series of Mach-Zehnder interferometers as well as ring and racetrack resonators, on a silicon-on-insulator platform were designed and fabricated for wavelength selectivity functions. In our design we used MMI couplers with AWG instead of the FPRs to overcome the common challenges such as mode mismatch at the FPR-waveguide array transition, significant phase errors, higher crosstalk and naturally larger footprint due to the increased number of waveguides for better resolution. Replacing the FPR region with MMI coupler makes the configuration more compact, uniformly distributes power among the waveguide array and results in lower loss and better mode matching due to the self-imaging property. AWGs are single mode optical waveguides which function as dispersive optical delay lines, a characteristic that makes them an ideal device for wavelength separation in WDM applications. However, AWG devices are subjected to various shortcomings particularly polarization-dependency, susceptibility to temperature change, and inevitable fabrication flaws [113-115]. These limitations can greatly affect the device's performance, predominantly the crosstalk from adjacent channels and phase delays owing to the non-uniformity in geometry dimensions resulted from sidewall roughness and fabrication imprecisions. In this work, a 4×4 MMI was designed and fabricated for the application of WDM. The 4×4 wavelength MMI demultiplexer was implemented by means of finding the optimal performance by altering the geometrical parameters of the MMI couplers. Advanced fabrication process was implemented and careful attention was paid to the design parameters of the AWG to mitigate the aforementioned limitations.

4.4.1 Device Operation

The WDM layout based on MMIs and AWGs is depicted in Figure 4-14.

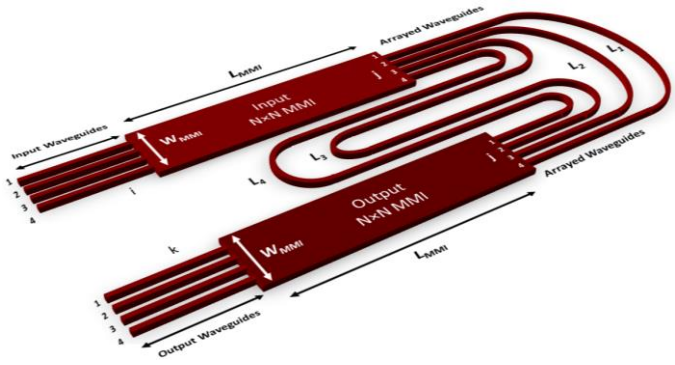


Figure 4-14: Illustration of an AWG-based multiplexer comprising two MMI couplers linked by an array of N waveguides.

As exemplified in Figure 4-14, the difference in length between the arrayed waveguides (ΔL) results in dissimilar bent sections which thus contribute to the occurrence of phase errors across the arrays. The operation principle of the AWG-based WDM is explained as follows. The system comprises three essential components: the input MMI which serves as a power splitter, the single mode arrayed waveguides, and the output MMI coupler which is the core element of the system. Light launched from any input access waveguide (input waveguides are denoted by i_N) of the 4x4 MMI splitter is reproduced at each output waveguide (output waveguides of the MMI splitter are denoted by j_N) in the form of multiple self-images with varying phase and uniform intensity [56]. The light then propagates through the array of waveguides before reaching the output MMI coupler. It is important to mention that the varying lengths of the arrays result in a phase shift in correspondence to ΔL. Therefore, the length of the single mode waveguides must be carefully calculated with the aim of enabling the optical fields to interfere in a manner that will attain de-multiplexing with optimal spectral efficiency and optical resolution, and low

loss, at the output of the MMI coupler (output waveguides of the MMI coupler are denoted by k_N). A further consideration, to ascertain low cross-talk and a symmetrical pattern of the multiple self-images, is the spacing of the arrays and their position at the MMI multimode waveguide. For that reason, the geometrical dimensions of the MMI coupler is equally critically important. In general, access waveguides are single mode strip waveguides, but the use of tapered waveguides at the input and output of the MMI waveguide is sometimes preferred. Tapered access waveguides serve to reduce back-reflections as well as the insertion loss and the crosstalk effect, also they are more tolerant to fabrication errors [58, 116].

The width of the MMI couplers was set to be wide enough to allow for adequate spacing between the arrayed waveguides while also ensuring uniform power splitting and minimized side-lobes in the spectral transmission. The length of both MMIs is calculated using Equation 15 [56], where, N is the number of input and output access waveguides, and L_π is the beat length which is defined by Equation 16.

$$L_{MMI} = \frac{3L_\pi}{N} \tag{15}$$

The beat length [117], is proportional to the square of the effective width of the MMI (W_e), which is the virtual width of the MMI waveguide which embraces the mode profiles evanescently decaying into the cladding.

$$L_\pi \approx \frac{4n_r W_e^2}{3\lambda_0} \tag{16}$$

where, n_r is the effective refractive index at the guiding region, and λ_0 is the center wavelength.

Finding the most optimum lengths of the array, is a fundamental challenge in AWG-based WDMs. In this work, the shortest or the reference arm of the array will be

signified by L_m, and for attaining compactness, the second arm was chosen to be the shortest. Setting the first or the last waveguide in the array as a reference arm would result in cross coupling and significantly longer arms, respectively. Moreover, in order to achieve nearly symmetrical pattern of the self-images, the reference arm selection is limited to L_2 or L_3. The radius of the waveguide bent regions in the arrays was also cautiously selected to circumvent immoderate radiation losses. It is not feasible to choose the lengths of the waveguides in the array to increase linearly along the array number j_N, hence, an array arm factor (d_j) must be introduced to aid in calculating the length of the arms adjacent to the reference waveguide as represented in Equation 17 [118].

$$L_j \approx L_m + d_j \Delta L \tag{17}$$

where, ΔL is the difference in length that yields a phase delay of $2\pi/N$ between the neighboring wavelength channels, and d_j is an integer number. The following shows the steps to calculate the length of the arrayed waveguides such that the respective phase delays of each arm along with the phase of the self-imaged output from the MMI splitter, result in a constructive interference at the output of the MMI coupler. The phase at the output of the MMI splitter j_N which is attributed by input i_N, is defined by [118]:

$$\varphi_{i,j} = \phi_1 - \frac{\pi}{2}(-1)^{i+j+N} + \frac{\pi}{4N}\left[i + j - i^2 - j^2 + (-1)^{i+j+N}\left(2ij - i - j + \frac{1}{2} \right) \right] \tag{18}$$

where, ϕ_1 is a constant phase. The phase difference between j_N and k_N (shown in Figure 4-14) with respect to the reference input which we define here as m, is calculated by [118]:

$$\Delta\varphi_{j,k} = \frac{\pi(-1)^N}{2N}\left\{\left[(-1)^{j+m}(j-\frac{1}{2})-(m-\frac{1}{2})\right]\left[(-1)^{k+m}(k-\frac{1}{2})-(m-\frac{1}{2})\right]\right\} - \frac{\pi(-1)^N}{2}\left\{\left[1-(-1)^{j+m}-(-1)^{k+m}+(-1)^{j+k}\right]\right\} \quad (19)$$

where $\Delta\varphi_{j,k}$ is at all times a multiple of $2\pi/N$, which is the phase shift separating the neighboring wavelength channels. Subsequently, the phase pertaining to the length of the array arms which compensates for the transfer phase of the MMI splitter is given by [118]:

$$\psi_{j,k} = (-(N/2\pi)\Delta\varphi_{j,k}) \bmod N \quad (20)$$

where, the modulus operator is used to restrict the result of $\Psi_{j,k}$ to a range between 0 and N-1.Finally the array arm factor is calculated using the following equation [118]:

$$d_j = \psi_{j,k} + B_j N \quad (21)$$

In Equation 21, the term B_j was added to ensure no intersection occurs between the adjacent waveguides in the array. B_j is simply an integer number and the smallest value to avoid waveguides intersecting is favored. In this work, the reference arm was selected to be the second arm, and by using Equation 19 and Equation 20, $\Psi_{j,k}$ was calculated for m=2, results of which are shown in Table 4.1.

Table 4.1: Calculated values of $\Psi_{j,k}$ for m=2 in a 4x4 MMI coupler.

	j_1	**j_2**	**j_3**	**j_4**
k_1	3	0	2	1
k_2	0	0	0	0
k_3	2	0	0	2
k_4	1	0	2	3

From Table 4.1, it is evident that in the case of output k_1, the forth arm (j_4) would intersect with the third arm (j_3), given that $\Psi_{4,1} < \Psi_{3,1}$, for that reason the term B_j in Equation 21 is used. In this case, B_4 is set to one and the calculated d_4 is 5. However, the objective is to find the best $\Psi_{j,k}$ values for m=2 that would prompt the smallest d_j values and thus shortest combination of the arm lengths of the arrays. The optimum values of d_j are shown in Table 4.2. The set of d_j values selected are for the case of output k_4 for which the calculated d_{max} is the smallest resulting in the shortest combination of arm length.

Table 4.2: Optimum values of d_j for the 4x4 AWG-based WDM.

	d_1	**d_2**	**d_3**	**d_4**	**d_{max}**
k_1	3	0	2	5	5
k_2	0	0	0	0	-
k_3	2	0	4	6	6
k_4	1	0	2	3	3

The reference arm was set to be 100 μm and length of other waveguides in the array was then calculated using Equation 17, in which ΔL is expressed as [118]:

$$\Delta L = \frac{\lambda_0 \left(\lambda_0 + \Delta\lambda \right)}{N \Delta\lambda n_g} \tag{22}$$

where, λ_0 is the center wavelength, $\Delta\lambda$ is the channel spacing which in this case is 100 GHz around 1310 nm, N is the number of access waveguides, and ng is the group index of the arrayed waveguides. Having calculated ΔL, the lengths of the array was then calculated using Equation 17, results of which are tabulated in Table 4.3.

Table 4.3: Calculated lengths of the arrayed waveguides with reference arm being the second arm.

Array #	Length (µm)
1	275.66
2	100.00
3	450.32
4	625.98

4.4.2 Simulated Device Performance

The prime objective of this work is to explore the operation of AWGs combined with MMI couplers, with WDM favored specifications, for optimal device performance. Specifications include low loss and crosstalk, and reduced channel spacing which is a crucial requirement in optical networks. This work is targeted to secure a 100 GHz channel spacing in 4-channel WDM, a 3-dB optical bandwidth of about 100 GHz, high uniformity and excess loss less than 1 dB for the MMI couplers. A considerable merit of our device is the Si layer thickness, which is minimized to only 0.161 µm and the Si waveguide width which is set to 0.35 µm. The small dimensions of the device necessitates a stringent control of the fabrication process for reliable performance. Compact MMI waveguide dimensions in which the modes are less confined and light spreads into the cladding layer, leads to a larger beat length and thus longer MMI couplers relative to the conventional thick waveguides. However, in our design, the width and length of the MMI section were optimized to achieve a compact device yet a high fidelity interference pattern at the output of the MMI couplers.

Lumerical MODE Solver was used to determine the optimal parameters and performance limitations of the 4×4 MMI and AWG-based WDM. The width of the

MMI splitter was set to be 6.2 μm for which the length was calculated to be around 78.62 μm. On account of the large dimensions of the MMI device, Variational FDTD (varFDTD) Solver was used, which requires less computational time than Lumerical 3D FDTD and produces results with similar accuracy. A comparison between 3D FDTD and varFDTD Solver results for the MMI splitter intensity at each output for excitation at input 2 and wavelength 1310 nm is shown in Table 4.4. The comparison clearly shows the strong resemblance between the 3D FDTD and varFDTD which in turn encouraged the use of the latter. The input and output waveguides, which extend into a taper, of the MMI couplers are strip waveguides with a width of 0.35 μm and a height of 0.161 μm. The tapers, with gradually varying width along the direction of propagation, were optimized to inhibit back reflections at the taper-MMI waveguide interface. The length of the tapers was carefully selected to circumvent mode mismatch between the guided modes in the taper and the modes supported by the MMI waveguide. Therefore, the length of the taper plays an important role in suppressing the total loss of the MMI couplers, but a trade-off between the optimal length and the device footprint must be met. The tapered waveguide was designed with the following parameters: 0.161 μm silicon layer thickness, length of 15 μm, and a maximum width or 1.2 μm. The gap between the tapered waveguides is about 0.38 μm, which is sufficient to avoid cross-coupling .The GDs layout of the device is shown in Figure 4-15 with four variations of the MMI couplers length to compensate for fabrication errors.

Table 4.4: Comparison between 3D FDTD and varFDTD solver normalized intensity at the outputs of the 4×4 MMI splitter for excitation at input 2 and wavelength 1310 nm.

Output Port #	varFDTD Intensity	FDTD Intensity
1	0.2378	0.2387
2	0.1987	0.1978
3	0.2153	0.1911
4	0.2076	0.2045

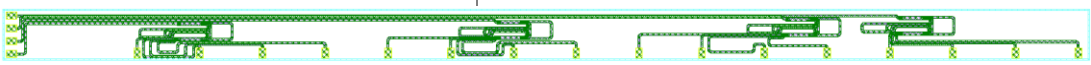


Figure 4-15: GDS layout of the four-channel AWG-based WDM where the shortest length is the second arm. Four variations of the MMI coupler length were designed to rectify fabrication errors.

It is relevant to note that the fabrication process used in this work, yields a sidewall angle of about 84 degrees. Therefore all elements were simulated with a sidewall angle of 84 degrees such that the desired width is at the center of the waveguide cross-section. Light was launched from the second input of the MMI splitter, realizing nearly symmetrical power splitting at the output ports of the MMI. The intensity pattern in the 4×4 MMI splitter is depicted in Figure 4-16, where the formation of the four images at each output port of the splitter at wavelength 1310 nm is conspicuous. The intensity at each output, shown in Table 4.4, indicates a good uniformity and an excess loss of ~ 0.658 dB for the 4×4 MMI splitter. The simulated phases at each output port of the 4×4 MMI splitter is tabulated in Table 4.5. The wavelength dependent response of the MMI splitter for launching at different inputs is shown in Figure 4-17: (a-d).

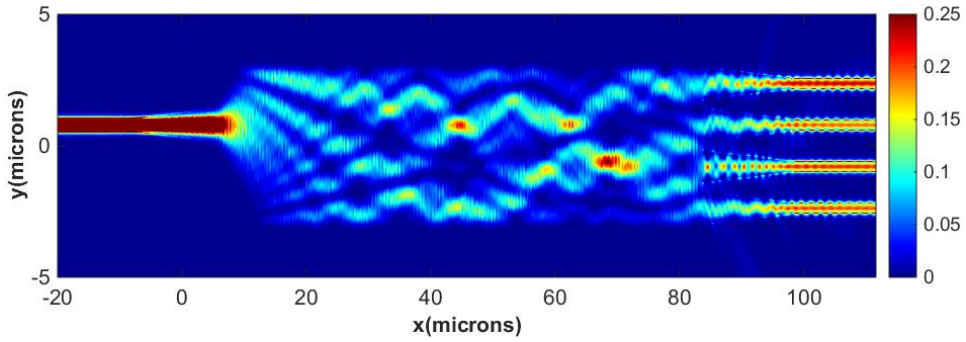


Figure 4-16: Intensity pattern in the 4×4 MMI splitter of length 78.62 µm, width 6.2 µm, and launching at input 2 for wavelength 1310 nm.

Table 4.5: Simulated phases at the outputs of the 4×4 MMI splitter for excitation at input 2 and wavelength 1310 nm.

Output Port #	varFDTD Phase
1	-2.1391
2	0
3	0.0341
4	1.3469

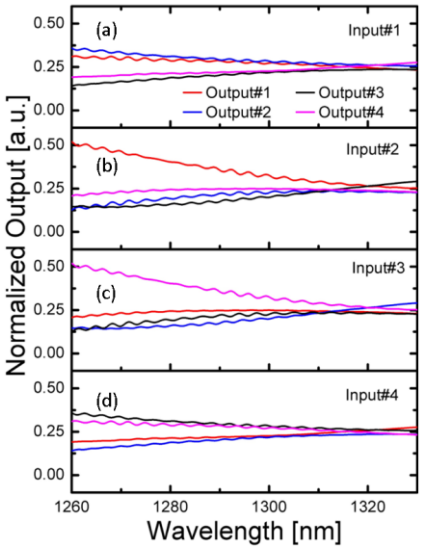


Figure 4-17: Wavelength dependent response of a 4×4 MMI splitter of length 78.62 µm, width 6.2 µm and launching at: (a) input 1 ; (b) input 2; (c) input 3; and (d) input 4.

Figure 4-17 shows a quite uniform pattern and an insignificant wavelength dependency near wavelength 1310 nm. The wavelengths being used in transmitting channels in the design of the AWG-based WDM, with a channel spacing of 100 GHz between adjacent channels, are around the 1310-nm region. The second MMI coupler was evaluated as follows. A mode source was placed at each input waveguide of the MMI coupler. The recorded intensities and phases from the output waveguides of the first MMI splitter were inserted as the amplitude and phase, respectively in the source at each input waveguide. Given the fact that the light is guided through the AWGs before reaching the second MMI coupler, the phase delay correlated with the different lengths of the AWGs must be accounted for in the simulation. The phase delays were addressed by introducing an offset to every source, which results in the launch of each source at different times. The corresponding offset was calculated using the following expression:

$$td = \frac{L_j}{v_g} \tag{23}$$

where, td is the offset, L_j is the length of array arms, and v_g is the group velocity of the single mode arrayed waveguides. The spectral transmission of the AWG-based WDM is demonstrated in Figure 4-18. Good uniformity and low side lobes is observed among the four different channels. Simulation results show that the channel spacing and the 3-dB optical bandwidth are around 100 GHz and that the device has a low crosstalk (~ -21 to ~ -25 dB), therefore, the device shows promising results and operates well on WDM technology.

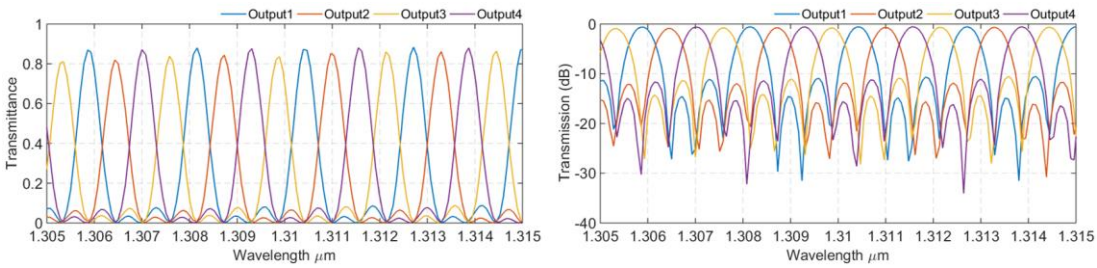


Figure 4-18: Spectral transmission of the four-channel WDM, with channel spacing of 100 GHz. The spectral response at all four outputs are shown for excitation at input 2 of the MMI splitter.

4.5 Summary

The principle of an unbalanced MZI gas sensor integrated with a temperature-compensating microring resonator was introduced and its operation was discussed thoroughly. Two different configurations were proposed: partially exposed MZI and suspended MZI gas sensors. Both configurations were unbalanced with respect to optical path length, yielding an interferometric response which is necessary for tracing the wavelength shift resulting from change in immediate surroundings. The partially exposed MZI sensor was simulated using MODE solver and a MATLAB script, results of which demonstrated a sensitivity as high as 2071 nm/RIU for TE mode. The suspended sensor, due to its complicated configuration (gratings used for mechanical support), was simulated using the S-parameters in 3D-FDTD together with the SPAR element Lumerical INTERCONNECT. Simulation results show a sensitivity reaching up to 5500 nm/RIU for TM mode, thereby establishing that suspended structures significantly enhanced the sensitivity due to greater modal overlap with the measurand. A temperature-compensating ring resonator was placed near the output of the sensors, in a manner that the output waveguide of the MZI sensor acts as the through waveguide of the temperature compensator. The ring resonator served in

estimating the temperature-dependent wavelength shifts which perturbs the function of the MZI gas sensor. This is achieved by not exposing the ring to the surrounding environment, consequently, any observed change in the response of the ring was merely due to the change in temperature within the waveguide.

A 4×4 AWG-MMI based WDM system was simulated using varFDTD due to the large footprint of the device. The device comprises two MMI couplers connected by an array of single-mode waveguides. The width of the MMI coupler was set to be 6.2 μm for which the length was calculated to be around 78.62 μm. An excess loss of around 0.658 dB was calculated when light was launched from the second input of the MMI coupler. The WDM system was designed to operate around the 1310 nm region with channel spacing of 100 GHz. A channel spacing and 3-dB optical bandwidth of around 100 GHz were achieved in the simulation with a crosstalk ranging from -21 to -25 dB. Therefore, the device shows a potential in operating as an integrated spectrometer.

Chapter 5: Device Fabrication and Optical Characterization

Following the fabrication of the MZI sensors, SEM and optical images were taken and are displayed in this chapter. Optical characterization results will be discussed after a brief description of the precise assembly of the optical and gas setups used for conducting the experiments under controlled conditions.

5.1 Fabrication

5.1.1 Partially exposed MZI sensor

The partially exposed MZI sensor was fabricated by A*STAR Institute of Microelectronics-Singapore via 248 nm optical lithography process with the aforementioned chosen dimensions: waveguide thickness of 220 nm, waveguide width of 450 nm, and Si slab height of 90 nm. The cross-sectional SEM image of the reference and sensor waveguide, of the fabricated partially exposed MZI sensor arms is shown in Figure 5-1.

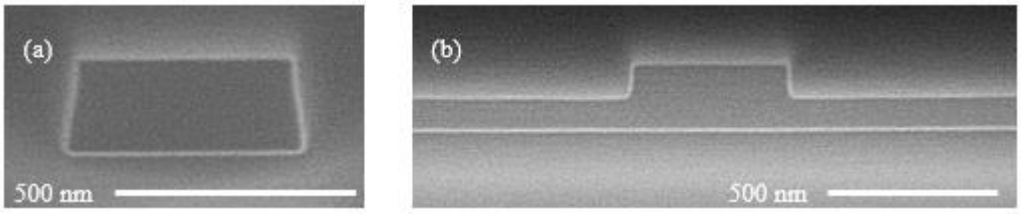


Figure 5-1: Cross-sectional SEM images of the fabricated partially exposed MZI sensor: (a) reference arm waveguide; and (b) sensor arm waveguide.

Optical image of the fabricated partially exposed MZI sensor is shown in Figure 5-2.

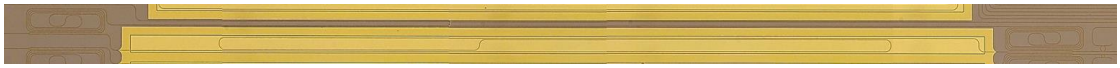


Figure 5-2: Optical image of the partially exposed MZI sensor with the temperature-compensating MRR near the output waveguide.

The dimensions of the fabricated sensor arms exhibited a sidewall angle which was measured to be around 85 degrees. The measured dimensions of the reference arm base width and height was measured to be about 465.1 nm and 230.8, respectively. The measured dimensions of the sensor arm base width and height was measured to be about 488.5 nm and 232.5 nm, respectively and a slab height of about 140.9 nm.

5.1.2 Suspended MZI sensor

The suspended MZI sensors were fabricated by Applied Nanotools (ANT) using a 100 keV electron beam lithography system. The optical image of the suspended MZI is shown in Figure 5-3.

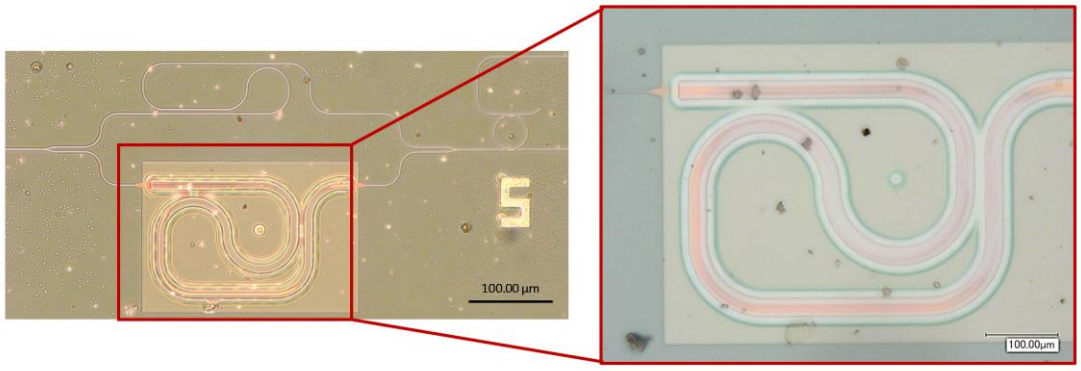


Figure 5-3: Optical image of the suspended MZI sensor with the temperature compensating MRR near the output waveguide.

A non-uniform color is observed in the serpentine pattern in the optical image of the suspended MZI sensor which indicates a non-uniformity in undercut.

SEM images of the sensor arm are shown in Figure 5-4 (a-c). It is clearly evident in Figure 5-4 (a) that the top cladding was not completely removed in the sensing region. Therefore, the propagating optical mode was not always exposed to the surrounding medium along the entire length of the sensing arm. In many regions the mode was isolated due to the presence of the top oxide cladding. The period of the gratings was designed to be 400 nm (200 nm Si region and 200 nm gap) for TM mode. However, it is apparent in Figure 5-4 (c) that the gap is much smaller than the Si fins. The fins were measured to be around 255 nm while the gaps were measured to be around 155 nm thus increasing the period by around 10 nm. The waveguide width was measured to be around 490 nm which is slightly larger than the design width. Therefore, given that the Si fin is now much larger than the gap, the effective index of the cladding is higher, resulting in a less confined mode which may lead to a substantial loss at the bent regions of the spiral sensing arm.

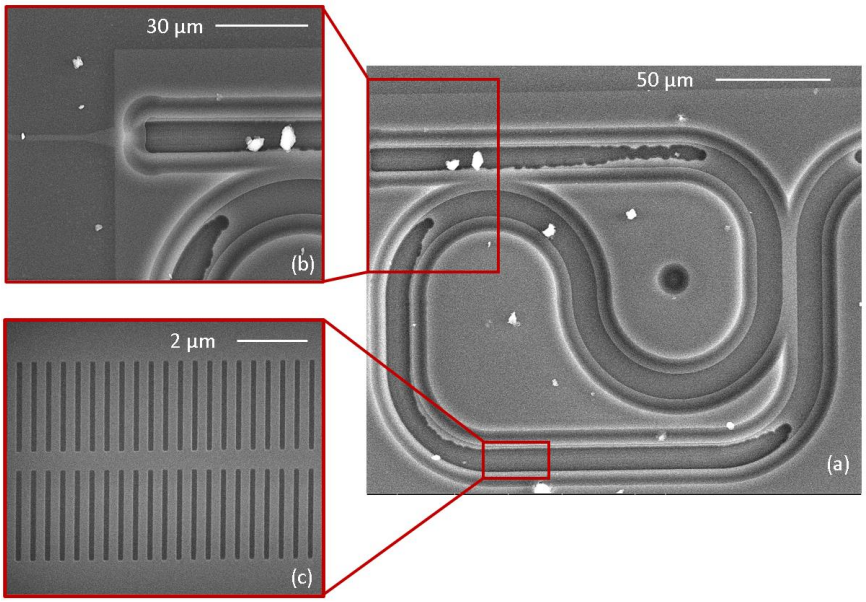


Figure 5-4: SEM images of the suspended MZI sensor: (a) Top-view image of the sensor arm; (b) Zoom-in image of the input region showing remains of the top cladding in the sensing region; and (c) Zoom-in image of the gratings.

5.2 Experimental Setup

5.2.1 Optical Characterization Setup

The optical test setup was comprised of a tunable laser source (Agilent 81600B) which has an output wavelength extending from 1460 – 1640 nm. A MATLAB script was employed to scan the laser wavelength for data collection. The wavelength sweeping time relies on several factors including the selected range of wavelength, the number of scans, the sweeping speed of the source, and the resolution. A tapered lensed fiber with a focal spot size of 1.7 μm was used to couple light from the laser source to the waveguides and from the waveguides to the optical power sensor (Agilent 81636B). The output was simultaneously imaged on a visible camera with a 20x microscope objective. An IR camera was also used to observe the light propagating along the waveguides. The optical chip was tightly secured on an aluminum chunk with the aid of a vacuum pump and was adjusted freely in the X, Y and Z directions by means of the Thorlabs piezo controller. The input and output fibers were also aligned using the Thorlabs controller. Light from the laser source is changed to any well-defined state of polarization using polarization controller Agilent 8169A. A picture of the experimental setup is depicted in Figure 5-5.

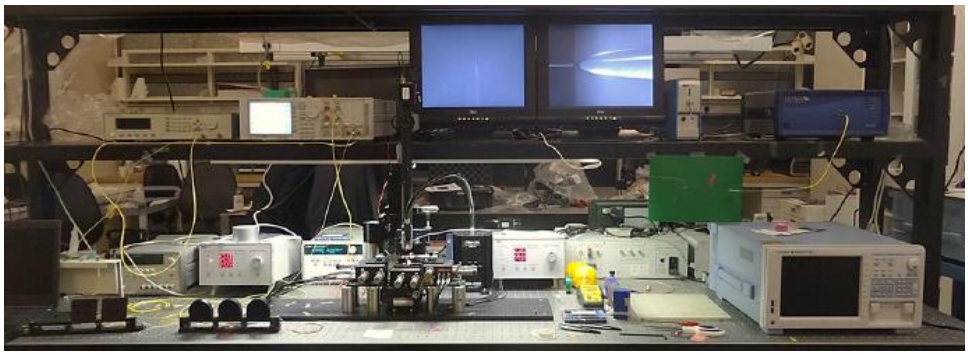


Figure 5-5: Schematic of the NOOR Lab Optical Test setup.

A separate optical setup was assembled for testing the suspended sensor since measurements were carried out by means of vertical coupling. The input and output grating couplers, designed in [119], were optimized to provide a maximum coupling efficiency for TM mode at an incident angle of 10°. Figure 5-6 shows the picture of the optical setup wherein the aluminum boats were customized for the angled fibers. The optical chip was placed on a translation stage for precise adjustments in the XYZ direction to optimally align the chip to the fibers. The fibers were probed at the surface of the chip with particular attention given to the gap between the tip of the fiber and the surface of the devices to be tested to minimize the insertion loss. Light from the tunable laser would first pass through a manual polarization controller to optimize TM mode before being coupled to and collected from the MZI sensor using input and output lensed fibers, respectively.

The optical chip was steadied on a vacuum chuck which was mounted over a rotator that enables an accurate alignment between the fibers oriented at a 10° angle and the chip. Lastly, the micro devices were viewed through the microscope with a working distance of 2.5 cm.

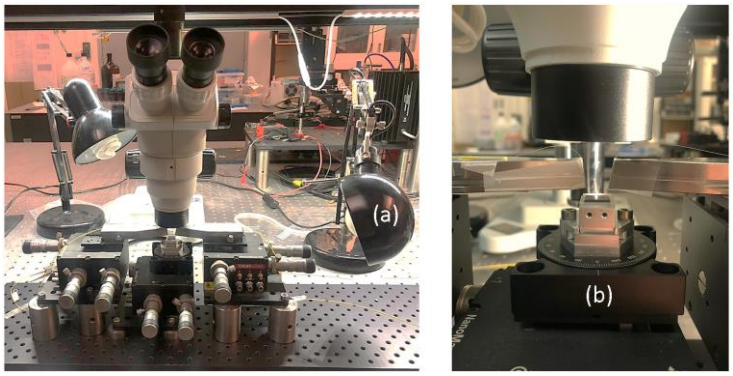


Figure 5-6: (a) Optical setup assembled to test the suspended MZI sensor by means of vertical grating couplers; (b) Zoom-in view of the boats customized for the fibers oriented at angle 10° for TM mode with a microscope for top-viewing the sensor.

5.2.2 Gas Chamber Setup

5.2.2.1 Partially Exposed MZI Sensor

As previously mentioned, the partially exposed MZI sensor was tested with He and N_2 gases alternatively introduced on to the optical chip, which was located in the gas chamber placed on the optical setup as demonstrated in Figure 5-7.

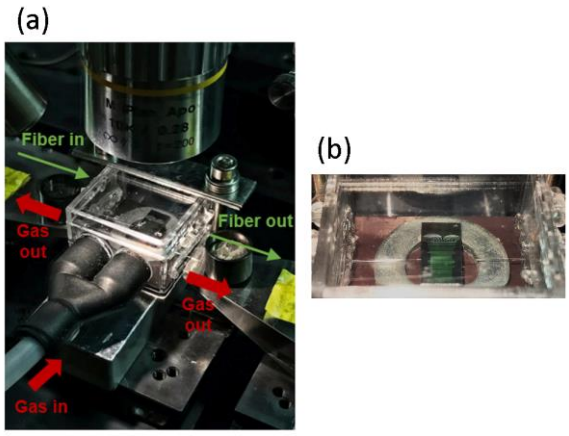


Figure 5-7: Gas chamber setup: (a) Photograph of the gas chamber on the optical setup; and (b) Close-up of the chip inside the gas chamber.

The chamber used comprises a plastic box with a plastic lid at the top and a gas inlet valve, through which the gases flow at a controlled flowrate into the chamber. The chamber has two narrow openings on either side, allowing access of the input and output lensed fibers to the edge of the optical chip, as shown in Figure 5-7. Each gas was purged from a calibrated gas cylinder, which is coupled to a mass flow controller (MFC), to accurately flow the test gases at a set rate, and then into the chamber. In an attempt to objectively quantify the sensitivity of the partially exposed MZI, both gases were flown at an equal rate of 0.408 SCFH (Standard Cubic Feet per Hour). Using the conversion factors in [120], to equate the flow rates of both gases, the MFC air flow

was set to 0.4 SCFH, and 0.15 SCFH at STP (Standard Temperature and Pressure) for N_2 and He, respectively.

5.2.2.2 Suspended MZI Sensor

For the suspended MZI gas sensor, given that light coupling is vertical, a different gas cell was designed using acrylic. The cell was engineered to assure a vacuum-sealed environment subsequent to bonding onto the optical chip by means of epoxy adhesives. The cell must be positioned in such a way that the horizontal sides of the chip remain exposed for easy, non-contact optical probing at the vertical grating couplers. Therefore, designing the gas cell's width to be thinner than the optical chip is a prerequisite for a precise control over fiber-grating coupler positioning.

In order to ensure that the suspended MZI sensor is completely sealed within the gas cell, a bonding technique was implemented to facilitate the alignment process. An acrylic-based body with a stopper for holding the chip (indicated by orange box) in place was designed and fabricated as shown in Figure 5-8 (a). A customized lid (shown in green in Figure 5-8 (b) was then used to cover one side of the chip where the input grating couplers are located and ensuring enough space for the thin gas cell to be positioned without overlapping with the position of the output grating couplers (Figure 5-8 (c)). Therefore, after aligning and fastening the gas cell with epoxy onto the center of the chip and removing the green lid, all vertical grating couplers are exposed while the sensing devices are fully encompassed within the cell.

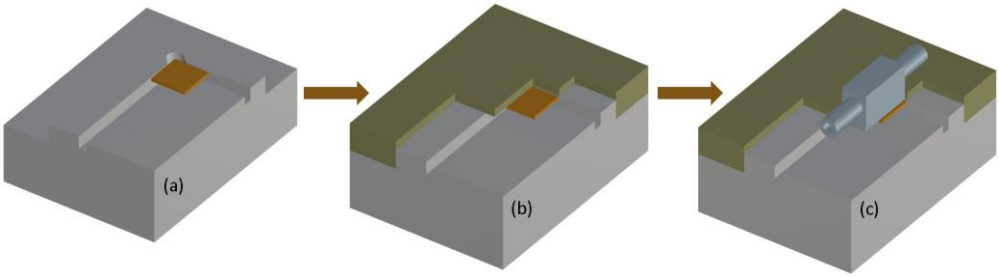


Figure 5-8: Acrylic-based body with a stopper for holding the chip and facilitating the chip-gas cell center alignment.

5.3 Experimental Results

5.3.1 Optical Characterization

5.3.1.1 Partially Exposed MZI Sensor

The spectral response of the partially exposed MZI sensor is depicted in Figure 5-9. The FSR was measured to be around 4.3 nm, which is well-matched with the simulated FSR. The measured fringe visibility reaches 22 dB around 1550 nm, easing the tracking of the fringes, due to the narrowing of the transmission dips. Therefore, obtaining a high interference pattern contrast is a vital requirement for a precise assessment of the sensor performance in real-time gas detection. The through response of the temperature-compensating MRR is also shown in Figure 5-9 with arrows.

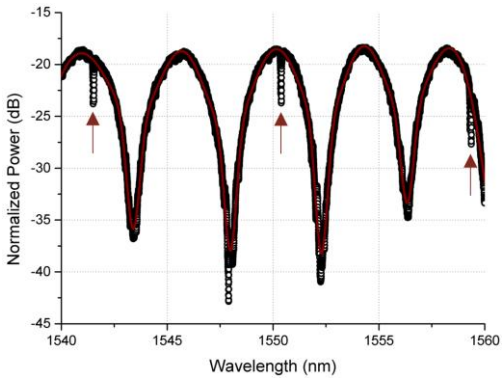


Figure 5-9: TE mode transmission spectrum of the MZI sensor (Fitting in solid red line), together with the through response of the MRR (indicated by red arrows).

5.3.1.2 Suspended MZI Sensor

Figure 5-10 shows the transmission, for TM mode, of a straight 450 x 220 nm^2 buried waveguide. The grating couplers exhibited a 3-dB bandwidth of around 51 nm.

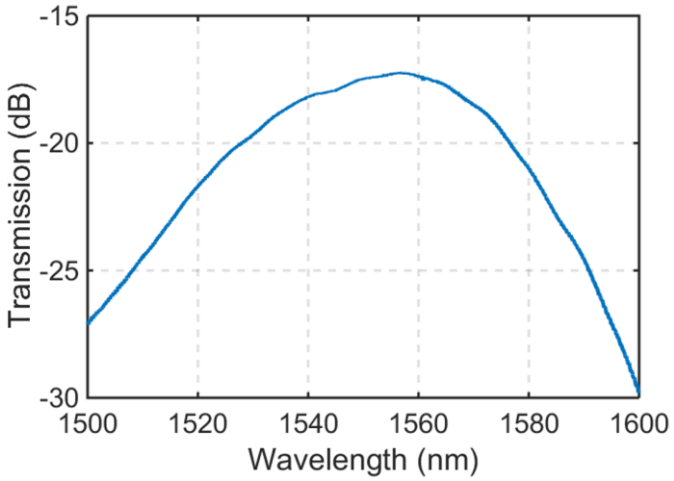


Figure 5-10: Measured transmission of a straight buried waveguide with an envelope curve profile of the grating coupler with a center wavelength of 1550 nm. The coupler exhibited a coupling loss of ~17 dB for TM mode and a 3-dB bandwidth of around 51 nm.

Figure 5-11 (a) shows the measured transmission of the suspended MZI sensor, however, no interferometric response was observed, only the through port response of the temperature-compensating ring resonator, which is located at the output of the MZI, was measured. This indicates that the loss at the sensor arm was significantly high due to the non-uniformity in the fabrication and the larger Si fins which increased the cladding index, reduced confinement, and thus increased bending loss. The ring response features an envelope curve resembling the transmission profile of the input and output vertical grating couplers. However, the suspended MZI sensing performance was not possible to determine due to the aforementioned fabrication

errors which lead to the absence of the MZI fringes. Figure 5-11 (b) shows the drop port response of the temperature-compensating ring resonator.

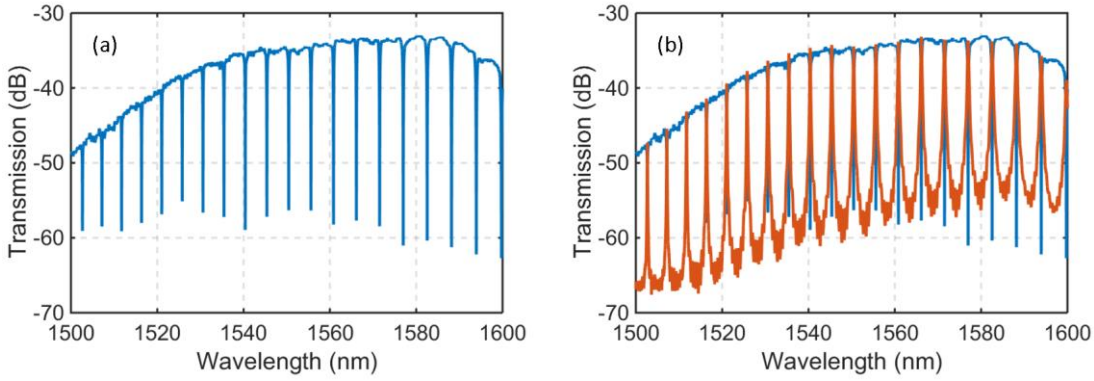


Figure 5-11: Measured response of the suspended MZI: (a) through port response of the temperature-compensating ring resonator, MZI fringes was not observed indicating a significant loss at the sensor arm; and (b) drop port response of the temperature-compensating ring resonator.

5.3.2 Gas Sensing Experiment

As mentioned earlier, the partially exposed MZI sensor and the temperature-compensating MRR were subjected to two different gases alternatively: He and N_2. Introduction of the different gases into the chamber results in a temperature drift which leads to a change in the core and the cladding refractive index. The variation in the refractive indices contributes to the change in the effective index which induces a wavelength shift. Therefore, subsequent to the exposure of different gases, a blue wavelength shift of about 36 pm was observed in the MRR response. The blue shift indicates that the temperature dropped when the gas in the chamber switched from He to N_2. Having measured the corresponding wavelength shift, using Equation 13, the change in the effective refractive index was calculated to be on the order of 10^{-5}. The calculated change in the effective refractive index was then added to the nominal value

of the effective index in order to get the new index value ($n_{eff_sen_new} = n_{eff_sen} + \Delta n_{eff_sen}$). Consequently, with the aid of MODE solver simulations, the corresponding refractive index value of Si was determined and thus the equivalent refractive index change. Finally, the TO coefficient of Si at 1550 nm was used to infer the drop in temperature which was then found to be about 0.27 K. Following the realization of the temperature drop, resulting from interchanging the gases, a similar approach was implemented for the MZI sensor. MODE solver was also used to examine how the sensor will behave in different temperatures and different gases. Different effective indices were obtained, which were then transferred to a Matlab script to get the resultant spectral shifts as shown in Figure 5-12. The plot demonstrates relative wavelength shifts in the presence of gases solely and simultaneously with taking the measured temperature change occurring after the switching of the gases into consideration. It was observed that the sensitivity of the partially exposed MZI sensor decreased by around 10.5% at a wavelength of 1552 nm, after considering the temperature effect. By this means, via the use of the ring resonator, the discrepancies in the spectrum wavelength shift due to inevitable temperature change, can then be calibrated for the MZI sensor.

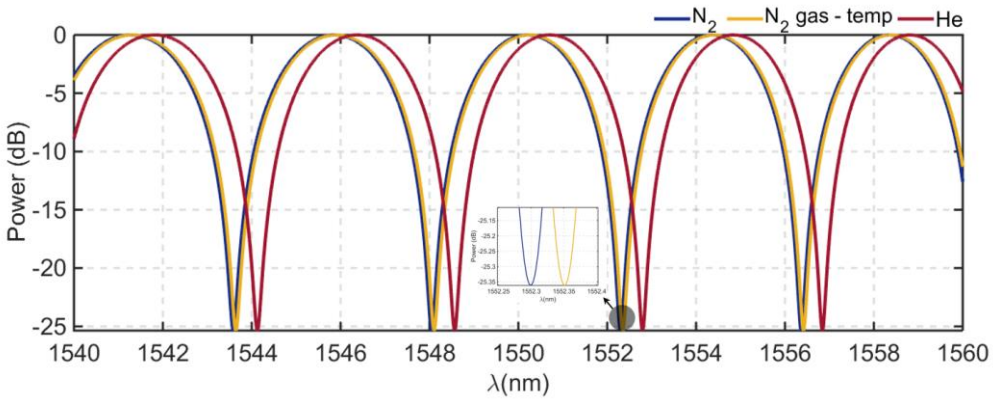


Figure 5-12: Simulated sensitivity of the partially exposed MZI, with taking the measured temperature drop into consideration. The zoom-in plot depicts the

wavelength shift in presence of N_2 alone (blue line) and the relative shift due to both: N_2 gas and temperature drop (yellow line).

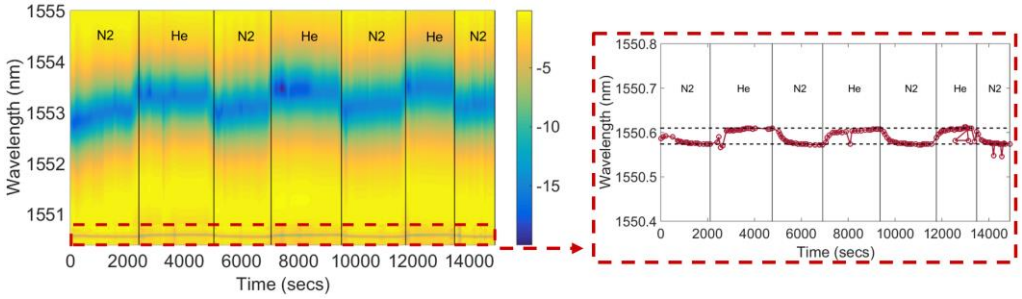


Figure 5-13: Wavelength shift resulted from switching the gases for the partially exposed MZI sensor. The color bar represents the transmission in dB scale. The surface plot (left) represents the response of the MZI and the MRR in presence of both gases: He and N_2. A blue wavelength shift of about 350 pm is observed in the MZI response due to gases and temperature change, while a blue wavelength shift of about 36 pm is detected in the MRR response (right) due to temperature drop alone.

Experimental results of the fabricated sensor are shown in Figure 5-13. The blue stripe represents the transmission dip near wavelength 1552 nm. The colorbar represents a 20 dB range used for plotting the optical power. A 350 pm shift towards the shorter wavelength was observed in the MZI response when the gases switched from He to N_2, yielding a high measured sensitivity of about 1458 nm/RIU. It is apparent from the MRR response that there is a change in temperature that influences the performance of the MZI sensor which elucidates the difference between the simulated and measured MZI gas sensor sensitivities. In addition to that, the 85 degree sidewall angle of the reference and sensor waveguides was not originally contemplated in the simulation model. The performance of the MZI gas sensor investigated in this work is summarized and compared with other optical gas sensors reported in the literature (Table 5.1).

Table 5.1: Comparative table of state-of the-art optical sensors for different gases.

Sensor Type	Sensor Configuration	Gas	Bulk Sensitivity (nm/RIU)	Limit of Detection (RIU)	Reference
Photonic Crystal (PhC)	PhC air slot cavity	Nitrogen (N_2), Helium(He), Carbon-dioxide(CO_2)	510	1×10^{-5}	[121]
	Slot PhC	N_2, He, CO_2	421	1×10^{-5}	[122]
Microcavity	Optofluidic Fabry-Perot (FP) resonator	-	1053	5×10^{-4}	[123]
	Fiber FP cavity	Ethylene (C_2H_4), Methane (CH_4), He, CO_2	812.5	1.2×10^{-6}	[124]
MZI	Partially exposed MZI	He, N_2	1276	9.7×10^{-5}	This work

5.3.3 Temperature Sensing Experiment

In the interest of resolving the challenges encountered due to the temperature drifts, leading to ambiguous measurement, a separate temperature sensing experiment was carried out for both, MZI and MRR sensors. The experiment was meticulously conducted by monitoring the dip of the transmission as the temperature of the chip was elevated from ambient temperature by about 10°C. This experiment was accomplished by putting the optical chip on top of a Peltier element. A thermocouple was secured on the surface of the chip for real-time temperature detection. A voltage was applied to the Peltier element and a respected time was maintained between every increment in temperature, ensuring stabilization of temperature. The temperature sensitivity response is shown in Figure 5-14. A red shift and a blue shift in the responses was observed for the temperature-compensating MRR and the partially exposed MZI, respectively, with increasing temperature.

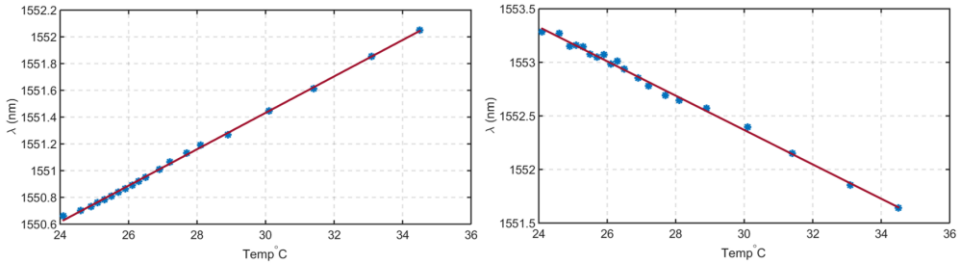


Figure 5-14: Temperature sensitivity of the temperature-compensating MRR (left) and the partially exposed MZI sensor (right) (with fit in solid red line).

The measured temperature sensitivity for the MRR was about 136 pm/°C which tallies well with the MRR temperature-induced wavelength shift which was ensued during the switching of the gases and the measured temperature drop as previously explained, yielding a sensitivity of about 133 pm/°C. The temperature sensitivity of the MZI which is determined by the slope of the right response in Figure 5-14 was measured to be around 166 pm/°C. The results indicate that the temperature sensitivity of the partially exposed MZI sensor is insignificant when compared with the sensitivity measured during the gas experiment which was around 1458 nm/RIU. The final stage to estimate the actual gas sensitivity, by excluding the temperature influence on the sensor, was reached by working through the following steps. The temperature sensitivity of the MZI (166 pm/°C) multiplied with the approximated temperature drop (0.27 K), provides the temperature dependent wavelength shift which was about 44.82 pm. Subtracting that value from the total wavelength shift measured during the gas sensing experiment (~350 pm), discloses the gas-induced shift alone. Therefore, the measured gas sensitivity with no temperature effect was measured to be around 1272 nm/RIU indicating a reduction in the sensitivity by about 12.5% which corresponds well with the expected sensitivity percentage drop. The MZI sensor also shows promising LOD, which in this work is defined by the 3-sigma deviation of the

probability density function of the distribution of the measurements. The measured LOD was 9.7×10^{-5} RIU, wherein the deviation was estimated to be 123.6 pm.

5.3.4 Liquid Sensing Experiment

A liquid sensing experiment was also conducted using Deionized (DI) water as reference and different concentrations of Isopropyl alcohol (IPA) as testing mediums. The experiment was conducted by bonding a microfluidic channel, made of PDMS, on top of the Si chip with the partially exposed MZI sensor. A microfluidic channel was used, as opposed to putting a drop of the solutions using a pipette, on the surface of the sensor for the purpose of ensuring stability and avoiding measurement ambiguity. A layout of the serpentine pattern of the fabricated channel is shown in Figure 5-15, the channel was aligned with the MZI sensor. Figure 5-15also shows an image of the fabricated microfluidic channel bonded onto the Si chip.

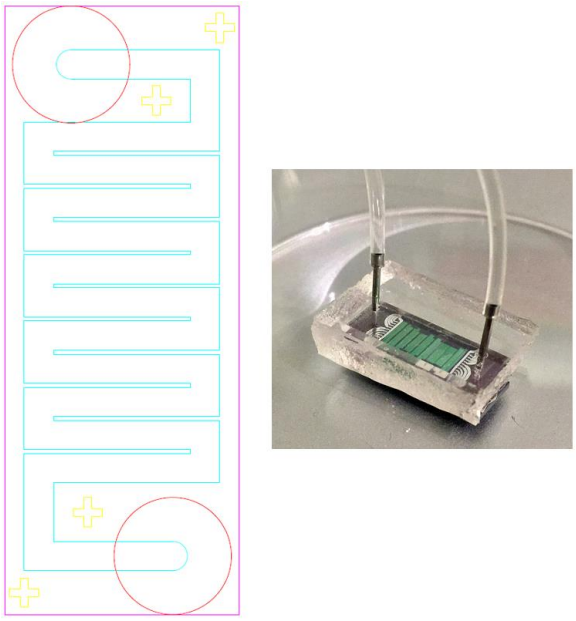


Figure 5-15: Layout of the serpentine-pattern of the microfluidic channel (left) and an image of the fabricated channel bonded on top of the Si chip (right).

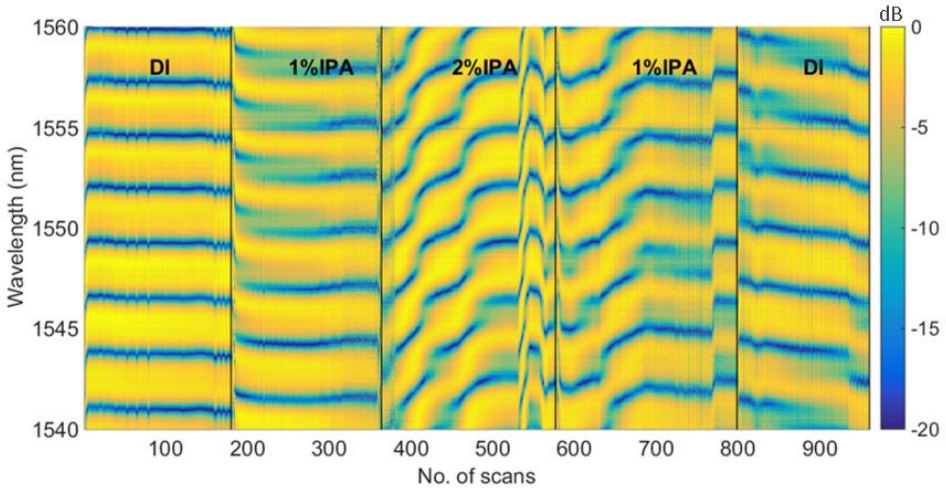


Figure 5-16: Surface plot representing the wavelength shift resulted from flowing different concentrations of IPA along the microfluidic channel.

Figure 5-16 shows the response of the partially exposed sensor to the varying concentrations of IPA. The results clearly demonstrate that the induced shift was larger than the FSR of the MZI sensor despite the small concentrations of IPA, hence it was difficult to quantify the sensitivity in this case. However the partially exposed MZI sensor can still be used for liquid sensing applications, by using even smaller concentration of IPA, indicating that the sensitivity of the sensor is very high.

5.4 AWG-MMI Spectrometric Results

GDS layouts of the AWG-MMI configuration with the optimum device geometries were prepared based on the GLOBALFOUNDRIES (GF) 45 CLO technology. The devices were fabricated at GF as part of the multi-project wafer (MPW) run for technology development. Grating couplers were attached to the input and output ends of the devices. Optical characterizations were conducted with the help of an O-band tunable laser source and a broadband photodetector. The light from the tunable laser is coupled to the input grating coupler through an inline-polarization controller. The polarization controller is used to align maximum input light to the TE polarization. At

the output of the device, light is collected using another grating coupler and fed to the photodetector. The laser and the photodetector are interfaced using LabVIEW to automatically acquire the wavelength dependent response of the device.

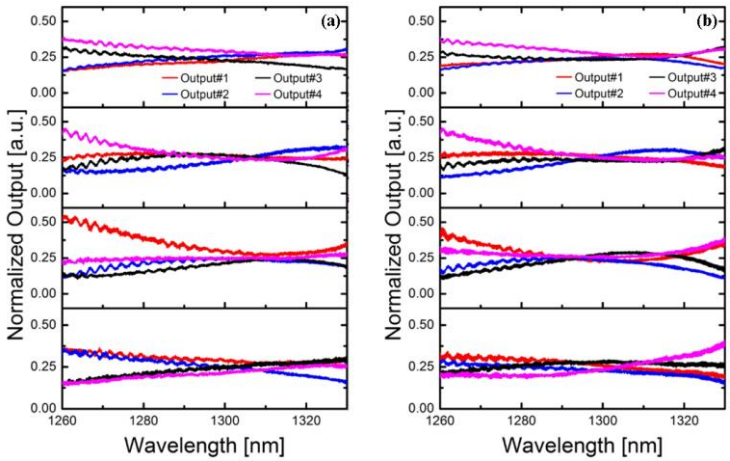


Figure 5-17: Wavelength dependent transmission characteristics measured at all the output ports for excitation at various input ports of the devices with: (a) Lmmi = 78.62 µm; and (b) Lmmi = 79.62 µm. MMI width is set to be 6.2 µm for both devices.

Figure 5-17 shows the measured wavelength dependent transmission characteristics of the 4X4 MMI splitters when various input ports are excited. Figure 5-17 (a) and (b) shows the characteristics for the MMI length of 78.62 µm and 79.62 µm, respectively. The width of the MMI is designed to be 6.2 µm for both devices. The transmission is normalized such that the total power (sum of all four output ports) is unity. The oscillations in the measured transmission curves could be due to the reflections which are also observed in the theoretical plots (Figure 4-17). Though a significant wavelength dependency is observed over the entire O-band, near to the desired wavelength (λ = 1310 nm) the splitting ratio remains close to 0.25 for both devices. This verifies the validity of the 2.5 FDTD simulation analysis of the devices. A small

wavelength shift is observed between the devices which is expected due to the slight difference in length.

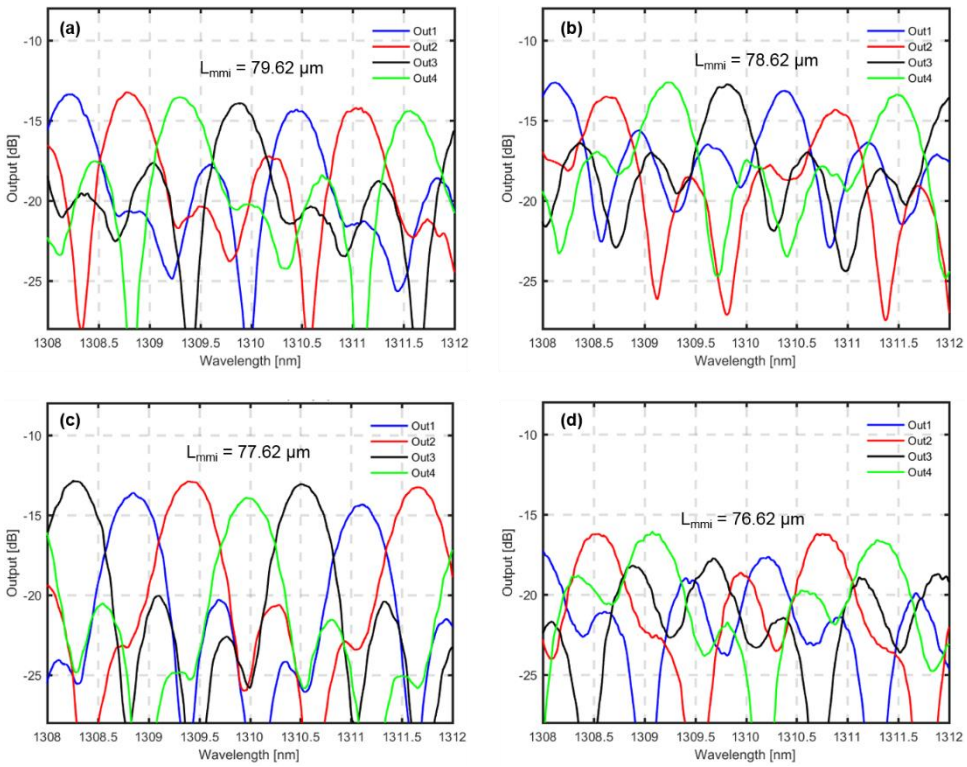


Figure 5-18: Transmission characteristics of arrayed waveguide gratings with: (a) LMMI = 79. 62 µm; (b) LMMI = 78.62 µm; (c) LMMI = 77.62 µm; and (d) LMMI = 76.62 µm, for excitation at input 2.

Figure 5-18 shows the transmission characteristics around 1310 nm for a four channel AWG with various MMI lengths for excitation at input 2. Four different designs were studied in this work with LMMI =79. 62 µm, 78.62 µm, 76.62 µm, and 76.62 µm for design#1 to design#4, respectively. The other AWG parameters including the MMI width (6.2 µm) and arrayed waveguide lengths were kept the same for all four devices. The measured spectrum includes the response from the grating couplers used at the input and the output. It is evident from Figure 5-18 that the output response of the AWG depends on the length of the MMI. Simulation results were found to be optimum

for the AWG with LMMI =78.62 µm; however, the experimental results show that the performance is better for the AWG with LMMI = 77.62 µm.

Measured AWG response is summarized in Table 5.2. The WDM channel spacing is designed to be 100 GHZ and the measured response shows the channel spacing varying between 88 GHz to 104 GHz. The slight variation in the channel spacing is attributed to the phase error in the waveguide array. Similarly, the channel bandwidth (3 dB) is measured to be 85 GHz – 99 GHz and the maximum variation is obtained for design#4 with LMMI = 76.62 µm. As mentioned earlier, best performance figures were measured for design#3 (LMMI = 77. 62 µm) with channel spacing 94 GHz – 102 GHz, bandwidth 87 GHz – 94 GHz and extinction ratio varying between 9 dB – 12.7 dB.

Table 5.2: Summary of the measured performance of the four devices near the desired wavelength λ = 1310 nm.

Device	*MMI Length [µm]*	*Ch. Separation [GHz]*	*3dB Bandwidth [GHz]*	*Extinction Ratio[dB]*
Design#1	79.62	90 - 104	86 - 87	6.2 - 7.8
Design#2	78.62	88 - 104	85 - 94	4.1 - 5.5
Design#3	77.62	94 - 102	87 - 94	9.0 -12.7
Design#4	76.62	94 - 101	85 - 99	4.1 - 4.6

The small variation between simulated and experimental results can be explained by the fact that the simulation was done in parts batches due to the large footprint of the device. Another potential reason could be related to deviations in the geometrical parameters of the fabricated devices which lead to phase errors. The deterioration of the extinction ratio was a result of the non-uniform splitting ratio at the output ports of the MMI splitter. As depicted in Figure 5-17, the splitting ratio of the MMI splitter with LMMI= 79.62 µm was nearly uniform at 1310 nm, whereas for the device with

LMMI= 78.62 µm, the splitting was uneven. This non-uniformity explains why the optical performance for the device with LMMI= 79.62 µm was better than that of Design#2 as shown in Table 5.2. Further optimization of the minimum waveguide separation parameter is needed in order to significantly reduce the channel crosstalk. Fine-tuning of the MMI splitter geometry and position of access waveguide can optimize the performance of the 4×4 MMI with regard to non-uniform power splitting.

5.5 Summary

A partially exposed MZI was fabricated by A*STAR Institute of Microelectronics-Singapore and demonstrated in this chapter. A gas sensing experiment was carried out using two gases: Helium and Nitrogen gases. A sensitivity of about 1458 nm/RIU was quantified and a limit of detection as low as 9.7×10^{-5} RIU was reached. The measured sensitivity was different from the simulated owing to the 85 degree sidewall angle of the fabricated waveguides as well as the temperature influence which was not accounted for in the simulations. Through the use of the ring resonator the temperature drift, occurring during the gas sensing experiment, was measured to be around 0.27 K. A separate temperature sensing experiment was conducted using the Peltier element on both devices: the MZI and the ring resonator, yielding sensitivities of 166 pm/°C and 133 pm/°C, respectively. Having calculated the temperature dependency of the MZI, the temperature dependent wavelength shift was then calculated and subtracted from the total measured wavelength shift. The MZI gas sensitivity was thus measured to be around 1272 nm/RIU which is 12.5% lower than the actual measured sensitivity.

A suspended MZI was fabricated by Applied Nanotools and demonstrated in this chapter. The SEM and optical images of the fabricated sensor revealed several errors which resulted in a substantial loss at the sensor arm and thus the device was no longer functioning as an interferometer and could not be tested for sensing.

The AWG spectrometer results are also shown in this chapter. Four different designs have been demonstrated with variations in the length of the MMI couplers used in the device configuration. The same width was used for all the designs which was set to be 6.2 µm. The MMI length for design#1 to design#4 was set to 79.62 µm, 78.62 µm, 77.62 µm, and 76.62 µm, respectively. Best optical performances were observed for design#3 where the channel spacing was ranging from 94 GHz-102 GHz, the bandwidth varied from 87 GHz – 94 GHz, and the extinction ration was measured to be 9 dB – 12.7 dB.

Chapter 6: Conclusions

For the purpose of realizing a sensor with optimal sensing performance with regards to high sensitivities and low detection limit, two MZI configurations were studied in this work. A partially exposed, 1-cm long MZI sensor and a suspended MZI sensor were designed and fabricated. The former, had a predicted sensitivity of around 2071 nm/RIU at a wavelength of 1550 nm, and temperature 300 K. An experiment was conducted with two different gases: He and N_2, which were alternatively introduced onto the optical chip, yielding a measured sensitivity of around 1458 nm/RIU. The difference in the measured and the simulated sensitivities stems from the fact that the temperature influence on the MZI resulting from swapping the gases was disregarded in the simulation. Another potential reason is the sidewall angle of the fabricated waveguides which was measured to be 85 degree. The angle is also expected to slightly vary along the arms of the 1 cm long MZI device.

In order to counteract the temperature effect, a ring resonator, buried in a 2 μm thick SiO_2 was positioned in a close proximity to the output waveguide of the MZI sensor. The response of the ring resonator helped determine the temperature variation happening during the experiment. Swapping the gases from He to N_2 caused a 0.27 K drop in temperature, which necessitated additional study to be conducted on the MZI sensor to eliminate the temperature-dependent wavelength shift.

Furthermore, independent temperature sensitivity test was performed on the MZI sensor together with the ring resonator temperature compensating sensor. The latter exhibited a measured sensitivity of 136 pm/°C whereas the MZI temperature sensitivity was measured to be 166 pm/°C. Fortunately the partially exposed MZI sensor is subjected to a significantly low temperature-dependency which is imperative

in optical sensing technology. The temperature-dependent wavelength shift was then withdrawn from the total wavelength shift, resulting in gas-dependent wavelength shift of around 305 nm and a sensitivity of 1272 nm/RIU. Therefore, the ring resonator helped determine the actual sensitivity of the MZI gas sensor which was identified to be 12.75% less than the actual total measured sensitivity which is in good agreement with the expected sensitivity percentage drop.

The suspended MZI sensor had a predicted sensitivity as high as 5500 nm/RIU at a wavelength of 1550 nm. Fabrication errors were observed in SEM and optical images, including non-uniform undercut, presence of top cladding in different regions in the sensing arm and grating with Si fins much larger than the design. Increasing the Si fin width increases the cladding refractive index which reduces the optical mode confinement and increases the propagation loss along the sensing arm as well as the bending loss. The aforementioned errors led to a considerable loss in the sensor arm and thus only the power from the reference arm was measured. Given that the temperature-compensating ring resonator was located at the output of the MZI sensor, after the y-combiner, its response was measured too. However, no sensing experiment was conducted on the suspended MZI sensor.

Finally in order to realize an integrated spectrometer, AWG-MMI based spectrometer was designed, fabricated and measured. The devices were fabricated at GF where the silicon single mode waveguide width was set to be 0.35 μm with a height of 0.161 μm. Despite the compactness of the device, high fidelity interference pattern and optimized optical performances were achieved. The MMI couplers had tapered input and output access waveguides to reduce the back reflections and thus the overall loss at the coupler output. The device was designed for a 100 GHz channel spacing in 4-channel WDM, a 3-dB optical bandwidth of about 100 GHz, high uniformity and excess loss

less than 1 dB for the MMI couplers. The simulated optimal length for the MMI was 78.62 μm while the optimal measured performances were for the device with MMI length of 77.62 μm. The reason behind the differences between the simulation and the experiment can be attributed to the fact that the simulation of the WDM system was not conducted as a whole but rather in batches due to the large footprint of the device. A low extinction ratio was achieved, resulting from the non-uniform splitting at the output of the MMI splitter which thereby urges the optimization of the minimum waveguide separation parameter. This parameter plays an important role in suppressing the crosstalk caused by the neighboring channels. In addition, small adjustments to the access waveguides orientation and the MMI waveguide dimension is required in order to achieve the desired optical performance. However, the AWG-MMI spectrometer designed in this work has the potential to function as an integrated spectrometer, eliminating the need for costly and bulky off-chip readout systems.

Chapter 7: Future Work: Microsystem-based Label-free Biosensing

For future work, we want to design the MZI sensor to operate in mid-IR range based on wavelength and intensity interrogation methods, for enhanced sensitivity and selectivity. For the integrated spectrometer, we want to optimize the waveguide separation parameter in the MMI in order to significantly reduce the channel crosstalk and enhance uniformity. For the bio-sensing topic, more simulations are to be conducted in order to better understand the behavior of the PMUT device in different mediums (air and water). The GDS has been sent to MEMSCAP (PiezoMUMPs process) for fabrication using UV-lithography. Subsequent to the arrival of the chips, the next step is to bond them into a microfluidic channel to allow for a controlled flow of the medium containing the cells. The final step involves bonding onto the optical chip, whereby the presented MZI sensor with chemically functionalized surface will be aligned with the channel. Therefore, after the cell lysis process is complete the released intracellular substances would bond to the immobilized receptors on the surface of the chip. The bonding between the receptors and the target molecules will induce a change in the refractive index at the surface which would result in a wavelength shift due to the change in the optical property of the propagating light by means of the evanescent field.

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